

The Macroeconomics of Firm Forecasts

Vladimir Asriyan Alexandre N. Kohlhas*

February, 2024

Abstract

We document a systematic increase in the forecasting ability of US firms over time and show that this increase is accounted for largely by the change in the firm-size distribution. We develop a macroeconomic framework of firm information production that is consistent with this evidence. We show that firms' size-dependent incentives to use 'data-driven decision-making' can rationalize the size-forecasting-ability relationship documented in the data. Consistent with the data, our framework implies that firms that use data more intensely allocate inputs more efficiently, adopt more efficient technologies, are more profitable and grow faster and larger. Our framework further suggests that data-driven decision making has important macroeconomic consequences: in a calibration exercise, we find that total factor productivity (household welfare) in the US would have been 8 percent (10 percent) lower in 2022 absent the increase in the forecasting ability of US firms over the past two decades.

Keywords: Information, forecasts, data economy, firm heterogeneity, increasing returns, strategic complementarities, amplification

*First draft: February 2024. Asriyan: ICREA, CREi, UPF, BSE (email: vasriyan@crei.cat). Kohlhas: University of Oxford (email: alexandre.kohlhas@economics.ox.ac.uk). We thank Timo Boppart, Jaume Ventura, Laura Veldkamp, Joshua Weiss, and seminar and conference participants at the AEA 2024, CREi, Oxford University, and ArmEA 2023 for their comments. Financial support from the Severo Ochoa Programme for Centres of Excellence in R&D (CEX2019-00915-S) and Handelsbank Stiftelsen is gratefully acknowledged. Erfan Ghofrani provided excellent research assistance.

1 Introduction

It is widely believed that advances in data-processing technologies have the potential to transform most economic interactions. Consistent with this view, over the past two decades, a sizable increase in the share of firms who systemically use data to inform their economic decisions has been observed. A simple estimate based on survey data from a sample of medium-to-large firms shows over a two-fold increase over the past ten years alone (Mckinsey and Co., 2023). Indeed, currently, around 30-40 percent of medium-to-large firms’ firms already employ some versions of *data-driven decision-making* (Brynjolfsson *et al.*, 2008).

Despite the increased use of data in firms, and the considerable resources devoted to its analyzes, the macroeconomic consequences of this rise in data-use are, however, not well understood. Should we expect the increased use of data to result in a more productive economy, because of the improved allocations of factors and increased productivity within firms? Have these changes in firms’ data use been large enough to affect economy-wide dynamics and business cycles? And should we expect the potential changes, and their macroeconomic consequences, to be mainly a short-run or long-run phenomena?

In this paper, we explore answers to these questions. We focus on one manifestation of advances in data-processing technologies for firms: its capacity to systematically improve firms’ economic forecasts. The rationale for this choice is both practical—as we can directly measure firm-level forecasts and connect them to firm-level choices and outcomes; and conceptual—since the seminal work of Lucas (1972), the role of information in improving economic forecasts has been shown central to understanding macroeconomic dynamics.¹

We argue that increases in firms’ forecasting abilities result in improved allocations of input, which increase overall productivity—both within and across firms. This is despite the fact that average firm size also increases in response. As firms’ forecasting performance improves, firms increasingly start to adopt riskier technologies, which further improves overall productivity, on average. Crucially, we demonstrate within the context of a standard heterogenous-firm model that the overall effects on the macroeconomy depend on the nature of strategic interactions. Our main contribution is to characterize these interactions between firms, their impact of firms’ information production choices, and how they evolve over time. In a quantitative exercise, we show that, due large complementarities in the adoption of data-processing technologies, total factor productivity (household welfare) in the US would have been 8 percent (10 percent) lower in 2022 absent the observe change in the forecast ability over the past two decades.

¹See, for example, Lorenzoni (2009); Blanchard *et al.* (2013); Chahrour and Jurado (2018); Angeletos and Lian (2016); Angeletos *et al.* (2021), among others.

Empirics: To empirically motivate our work, we use micro data on managerial forecasts from the I/B/E/S-Compustat panel. Using the merged firm-level data set, we document a systematic increase in the forecasting ability of US firms. Over the past two decades, managerial forecasts of one-year-ahead revenue, profits, and capital expenditures have all witnessed accuracy improvements between 20-50 percent, and similar developments are seen across most sectors. Importantly we document that, in the survey data, this increase in forecasts accuracy is tightly tied to developments in firm size. There is substantial heterogeneity in information use across the firm-size distribution.

Using a variety of estimation methods and controls, we find that larger firms, all else equal, produce more accurate forecasts. Indeed, in a simple simulation exercise we show that all of the observed increase in forecast accuracy over the past two decades can be attributable to the size-forecasting-ability relationship. Over time, firms have become larger and larger firms are associated with better forecasts. Technological improvements, changes in sectoral compositions, or changes in the volatility of variables, by contrast, appear to only have a minor influence on our results. (Comfortingly, we find similar estimates in the Duke-CFO survey, which features smaller firms than those in our sample.)

We next turn to cross-sectional differences in firm-level outcomes. We document that, all else equal, more accurate firms feature less misallocation, have higher levels of productivity, and are more profitable. We also find that more accurate firms grow faster and larger. Consistent with the view that these differences are driven by investments into data analysis and technology, we document that more accurate firms also spend more on R&D, software, and information management. Finally, we demonstrate that firms that acquire another firm, and as a result grow larger in size, see improvements to the accuracy of their forecasts over the next five years. We conduct a battery of robustness tests, which show that our results are resilient to issues related to alternative measures and specifications, and robust to the inclusion of firm-level controls. Combined with our earlier evidence, these results discipline the potential channel by which improvements in firms' accuracy have macroeconomic consequence.

Model: To explore the macroeconomic consequences of these developments, we embed firm information production into a workhorse heterogeneous-firm model. In our framework, firms choose whether or not to produce information subject to a fixed cost. The production of information has two benefits to the firm. First, it, all else equal, helps firms better predict future productivity, and thus is useful for determining the allocation of inputs. Second, it, all else equal, helps firms better decide on which products to produce-information allows firms to tailor their product to the tastes of the end consumer. Consistent with our estimates, our first main theoretical result demonstrates that firms which use information more intensely

allocate inputs more efficiently, adopt more efficient technologies, are more profitable, and grow faster and larger.

Our second main result concerns the possible macroeconomic consequences of a decline in the cost of information production. As illustrated in [Bloom *et al.* \(2019\)](#), over the past two decades, there has been a large decline in the cost of information processing. We show that the effects on the macroeconomy of such a decline depend critically on the nature of strategic interactions—in particular, whether firms’ input choices are strategic complements or substitutes. As time progresses and more inputs become variable, we show that strategic complementaries, all else equal, tend to dominate. This leads to further amplification of firms’ partial equilibrium responses to the decline in information costs. We characterize these effects, how they depend on the extent of firm heterogeneity, and show to decompose the overall effects into their partial equilibrium and general equilibrium components. We, furthermore, show that the general equilibrium amplification of a decline in information costs can be so large so as to result in the presence of multiple equilibria.

Finally, in a calibration exercise, we demonstrate that firms’ size-dependent incentives to use information-driven decision-making can rationalize the magnitudes of the size-forecasting-ability relationship observed in the survey data. For plausible parameter values, consistent with the observed increase in forecast performance, we document that total factor productivity (household welfare) in the US would have been 10 percent (8 percent) lower in 2022 absent the observed increase in the forecasting ability of US firms. We further decompose this overall estimate into a short-run and a long-run component. On balance, we find that that most (around 2/3) of the benefits accrue over the longer horizon, especially as firms start to adopt more efficient technologies and the increase in starts to firm size materialize.

Our results suggest that a large share of the macroeconomic consequences of information-processing arise due to general equilibrium interactions between firms. Beyond the analysis in this paper, we see important scope for extending our framework to also account for how advances in data technologies alter the way households decide which goods to purchase, and thereby their interactions with firms. Another avenue for future research is to combine models of optimal information production with those in which increases in firm size have detrimental welfare consequences, due to decreases in firm competition ([Edmond *et al.*, 2023](#)). The latter approach would allow for a more granular welfare assessment of the recent change in the firm-size distribution—one which separates the potential costs from the potential benefits. Overall, we view the research in this paper as a useful step towards a macroeconomic framework that accounts for the ongoing changes in data-processing technologies based on a minimal set of departures from workhorse macroeconomic models.

2 Motivating Evidence

2.1 Accuracy and Size

We present new evidence on the relationship between firm size and the accuracy of firm expectations. To start, we use micro data on firm forecasts from the I/B/E/S managerial guidance database. The I/B/E/S data set contains, for an individual firm-year, a manager’s publicly stated expectation for their firm’s revenue, profits, and other performance variables over the next year. We exploit one-year-ahead forecasts made concurrently with the release of the previous year’s financials. We link the I/B/E/S database to Compustat, which provides detailed data on firms’ financials. The merged I/B/E/S-Compustat sample covers the period 2002-2022. Appendix A.2 provides more information on the sample construction.

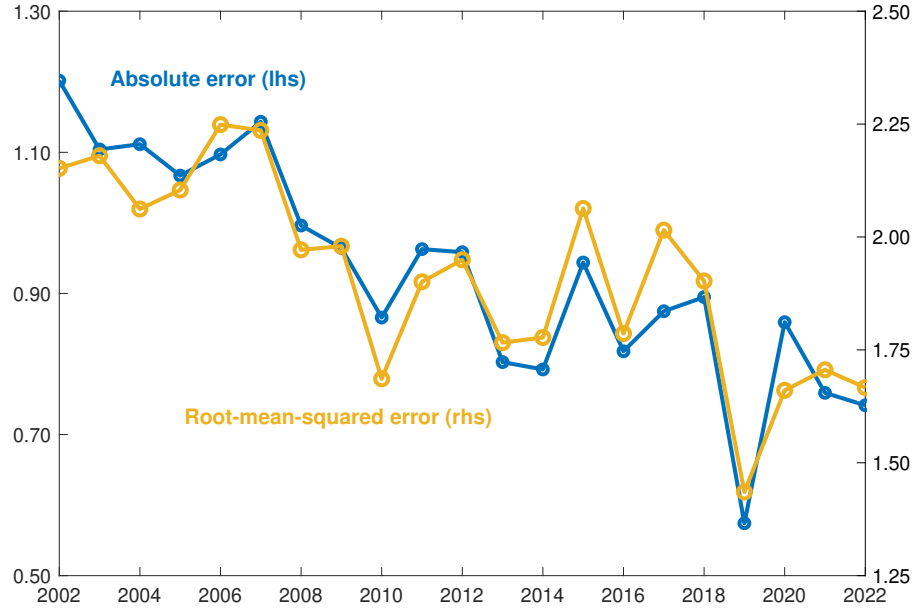
We begin by documenting changes to the accuracy of firms’ expectations and size over time. We focus on one-year-ahead revenue errors, defined as the realization minus its predicted value. A negative error thus corresponds to an over-estimate of future revenue. Panel a in Figure 1 showcases the average accuracy of firms’ expectations over time. All else equal, over the past two decades, firms’ expectations have improved considerably, with an average decline in one-year-ahead uncertainty between 24-38 percent, depending on the measure used. The increased accuracy of firms’ expectations is indicative of changes to firm characteristics, driving changes in their forecasting practices. Panel b in Figure 1 depicts, for the same period, changes in the firm-size distribution, measured by the number of employees. As documented in Autor *et al.* (2020) and elsewhere, over the past two decades, firm size has increased drastically—with, for example, a close to doubling in the share of firms with employment exceeding the 80th percentile of the 2002-employment distribution. Table A.2 in the Appendix confirms our initial findings using regressions of individual firm errors on time and firm size, respectively.

The simultaneous trends in the accuracy of firm’ expectations and firm size, combined with a sizable fixed-cost component to the processing of information (e.g., Brynjolfsson *et al.* (2008); Bloom *et al.* (2019)), raises the question of whether there is a relationship between the two trends. To investigate this question, Panel a in Figure 2 plots the difference between the average accuracy of one-year-ahead log-revenue forecasts within size (employment) quintiles and the overall average taken across all firm sizes. The results in Panel a in Figure 2 show a marked, monotone relationship between firm size and the accuracy of firm expectations in the raw data. All else equal, larger firms produce more accurate forecasts—with an especially pronounced difference when moving away from the bottom quintile of the firm size distribution.

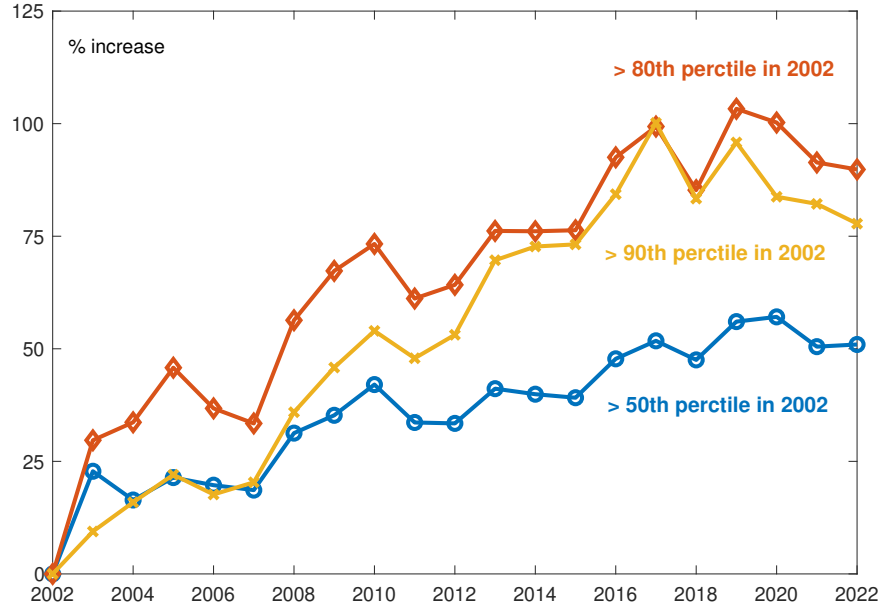
The relationship in Panel a in Figure 2 may be contaminated by other factors, such as firm age or differences in the volatility of revenues across firm size, that can simultaneously affect firm size and the accuracy of expectations. To address this issue, Panel b in Figure 2 plots

Figure 1: Time Evolution of Uncertainty and Size

Panel a: Uncertainty and Time

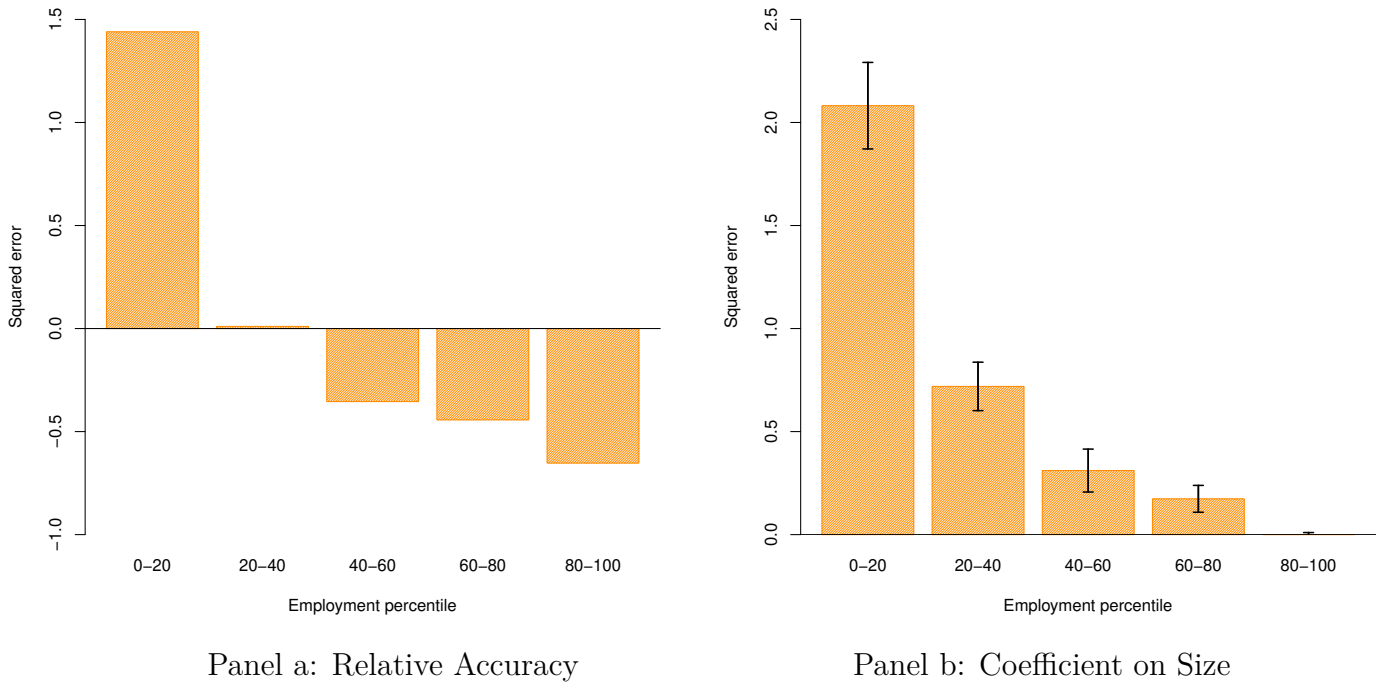


Panel b: Size and Time



Note: Data from the I/B/E/S-Compustat sample. Panel a: the mean absolute error of one-year-ahead revenue forecasts on the left vertical axis, and the root mean-squared-error on the right axis. Revenue errors are scaled by a firm's tangible capital stock. Panel b: the percentage increase from 2002 in the share of firms in a given year with employment exceeding the x th percentile of the 2002-employment distribution. The 50th, 80th, 90th percentile of the 2002-employment distribution correspond to around 1,000, 7,000, and 18,000 employees, respectively. Table A.2 in the Appendix shows the associated regression results.

Figure 2: Revenue Expectations Across the Size Distribution



Note: Panel a plots the difference between the squared error of one-year-ahead log-revenue forecasts from the I/B/E/S-Compustat merger within size (employment) quintiles and the overall average taken across all size levels. Panel b plots the coefficient estimates on size from a regression of the squared value of individual errors on the size quintile the firm belongs to, controlling for firm characteristics (Table I Column 3). Revenue errors are scaled by a firm's tangible capital stock and normalized by their mean value in the sample. Whisker-intervals are one-standard deviation robust (clustered) confidence bounds. Sample: 2002-2022.

the estimates from a regression of the accuracy of firm expectations on the firm-size decile, controlling for firm characteristics and time and sector fixed effects. Table I explores the effects of alternative controls and estimation methods. Our results confirm the relationship in the raw data. The accuracy of firm expectations decline monotonically in size, even after controlling for firm characteristics. A larger firm, all else equal, produces more accurate forecasts. The magnitude of the estimated effect of size is, furthermore, considerable: increasing the size of a firm by one quintile decreases the associated squared forecast error by 42 percent of its average value (Column 4 in Table I). Crucially, after controlling for size, the effect of time itself becomes insignificant (Columns 2 in Table I). This suggests that accuracy increases unrelated to size have played only a minor role over this period. Indeed, Figure A.10 in the Appendix, using estimates from Table I, shows that the change in the firm-size distribution alone can account for 75 percent of the observed increase in accuracy. With the addition of firm assets and revenue, as additional measures of size, this share increases to close to 85 percent.

We obtain similar estimates to those in Figure 1 and 2 for alternative measures of size and the accuracy of expectations. Table A.3 and Figure A.8 show similar estimates when proxying firm size with firm assets instead employment. Table A.3 documents that alternative measure of accuracy (e.g., the absolute error) likewise monotonically increase in size, irrespective of whether size is measured by employment or assets. We further find that across all-but-one sector accuracy has improved over time and increases with size (Figure A.9).

Table A.3-A.9 in the Appendix contain additional analysis and further robustness checks. We show that our findings extend to forecasts of other variables than revenue (firm profits and capex expenditures), to measures of the percentage increase in revenue, and to an alternative data set that measures the accuracy of firm forecasts of macroeconomic outcomes for smaller firms (*The CFO survey*; Graham *et al.*, 2023). We also show that the accuracy of firm forecasts improves after large acquisitions of other firms (acquisitions that are at least 10 percent of firm assets), as we all extend to different assumptions about sectoral and time fixed effects (Table A.5). The latter is important as the evidence in Table A.5 documents that the bulk of the revenue uncertainty faced by firms pertains to individual-specific shocks (rather than sectoral or aggregate).

In summary, the results in this subsection provide evidence for systematic heterogeneity in the accuracy of firm expectations. The growth in firm size over the past two decades have been accompanied by an associated increase in the accuracy of firm expectations, as larger firms produce more accurate forecasts. The data clearly reject the common-expectation assumption embedded in the full-information rational-expectation framework. Motivated by these findings, in the next section, we develop a heterogeneous-firm model with heterogeneity in the production information, consistent with this evidence. Before doing so, we, however, briefly

Table I: Revenue Expectations, Firm Size, and Time

	<i>Squared (log.) revenue errors</i>				
	(1)	(2)	(3)	(4)	(5)
Firm size	−0.468*** (0.055)	−0.454*** (0.052)		−0.423** (0.135)	−0.430* (0.226)
Firm size (1)			2.082*** (0.210)		
Firm size (2)			0.719*** (0.118)		
Firm size (3)			0.311*** (0.104)		
Firm size (4)			0.174*** (0.065)		
Time		0.007 (0.007)			0.117*** (0.028)
Firm age		−0.063** (0.032)	−0.073** (0.059)	0.143** (0.063)	0.117 (0.093)
Log rev. volatility				0.013* (0.007)	
Observations	12,488	12,488	12,488	6,817	2,569
Sector FE	✓	✓	✓	×	×
Firm FE	×	×	×	✓	✓
Time FE	×	×	✓	✓	✓
Panel GMM	×	×	×	×	✓
F statistic	3.911***	3.922***	4.321***	9.240***	NA.

Notes: Panel estimates from the merged I/B/E/S-Compustat sample. Column (1) shows estimates of the squared value of one-year-ahead log-revenue errors on firm size (employment) and sector (NAICS-4) fixed effects. Firm size is measured by the quintile the firm's employment is at time t relative to the 2002-employment distribution. Column (2) adds time and age controls. Column (3) allows for separate coefficients on size-levels (estimates are relative to the largest firms, those in the 80-100th percentile) and time fixed effects. Column (4) allows for firm fixed effects instead of sector fixed effects and controls for the individual four-year-rolling revenue volatility. Finally, Column (5) provides Arellano-Bond estimates. Revenue errors are scaled by firm capital and normalized by the overall average absolute (squared) error. The top and bottom 1 percent of errors have been removed. Robust (clustered) standard errors in parentheses. Sample: 2002-2022.

turn to documenting the extent to which firm productivity and profitability differ between more and less accurate firms. This will help latter corroborate several model predictions.

2.2 Accuracy and Firm-level Outcomes

We start by documenting systematic differences in sectoral misallocation. The misallocation of factors is one mechanism through which errors in firms' expectations manifest at the firm and economy-wide level (David *et al.*, 2016 and David and Venkateswaran, 2019).

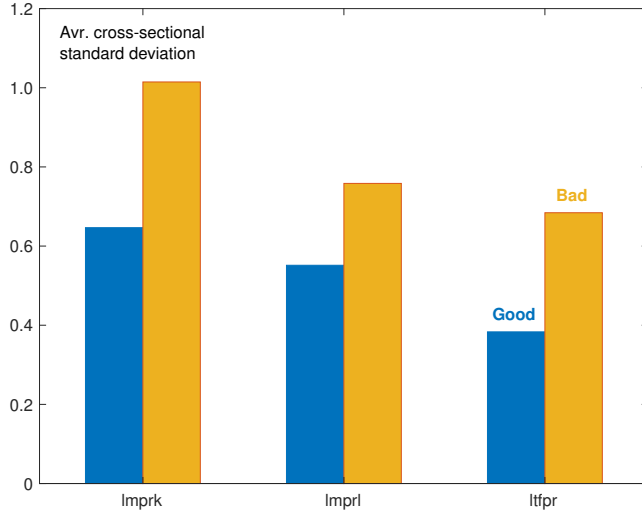
We define sectors by their four-digit NAICS industry classification. Building on the framework developed by Hsieh and Klenow (2009) and Gopinath *et al.* (2017), we compute our measures assuming a Cobb-Douglas production technology and monopolistic competition with CES demand. The profit-maximizing choice of an input for firm $i = \{1, 2, \dots, N_s\}$ in sector $s = \{1, 2, \dots, S\}$ at time $t = \{1, 2, \dots\}$, thus, equates its marginal revenue product with its (sector-specific) cost. We assume the presence of two factors of productions, capital and labor. As a baseline and for comparability with our model below, we set the labor share α equal to $2/3$, corresponding to the average labor share in the U.S. Our measures of misallocation are, however, not affected by the assumption that α is common across sectors, as these measures exploit within-sector variation of firm-level outcomes. Following the terminology in Foster *et al.* (2008), we define *revenue-based total factor productivity* (TFPR) as revenue divided by output net of firm total factor productivity (TFP).² The *marginal revenue product of labor and capital* (MRPL and MPRK, respectively) are, by contrast, defined as revenue divided by labor and capital employed by the firm, respectively. We take revenue, labor, and capital stock measures from the I/B/E/S-Compustat sample (Appendix A.2). Panel a in Figure 3 reports the average cross-sectional dispersion in $\log(\text{MRPL})$, $\log(\text{MRPK})$, and $\log(\text{TFPR})$.³ We report these estimates separately for accurate and inaccurate firms. We define “a good forecasting” firm as one that (i) is below the median in terms of the mean-squared-error of one-year-ahead log-revenue forecast; and (ii) one for which we have at least five observations.

Across the three different measures, Panel a in Figure 3 documents systematic differences between accurate and inaccurate firms in sectoral misallocation. Panel a shows that, all

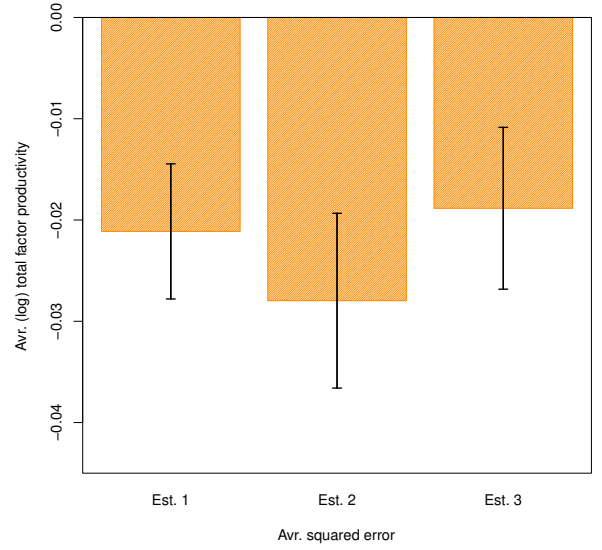
²The production technology used by firm i is $y_{i,s,t} = A_{i,s,t} \cdot l_{i,s,t}^\alpha \cdot k_{i,s,t}^{1-\alpha}$, $\alpha \in (0,1)$, where $y_{i,s,t}$ is firm output, $l_{i,s,t}$ and $k_{i,s,t}$ the amount of labor and capital employed, respectively, and $A_{i,s,t}$ is firm total factor productivity. Let $p_{i,s,t}$ be the firm-specific product price. Then *revenue-based total factor productivity* is defined as: $\text{TFPR}_{i,s,t} \equiv \frac{p_{i,s,t} \cdot y_{i,s,t}}{l_{i,s,t}^\alpha \cdot k_{i,s,t}^{1-\alpha}}$. The marginal revenue product of labor and capital are, by contrast: $\text{MRPL}_{i,s,t} \equiv \kappa_L \cdot \frac{p_{i,s,t} \cdot y_{i,s,t}}{l_{i,s,t}}$ and $\text{MRPK}_{i,s,t} \equiv \kappa_K \cdot \frac{p_{i,s,t} \cdot y_{i,s,t}}{k_{i,s,t}}$, where $\kappa_L \in \mathbb{R}_+$ and $\kappa_K \in \mathbb{R}_+$ are common constants that depend on the elasticity of substitution and the labor share and capital share, respectively.

³We compute cross-sectional dispersion measures in two steps. First, we compute the standard deviation across firms i in a given sector s and year t . Second, for each year, we measure dispersion for the whole economy as the weighted average of dispersions across sectors. We give each sector a time-invariant weight equal to its average share in overall employment.

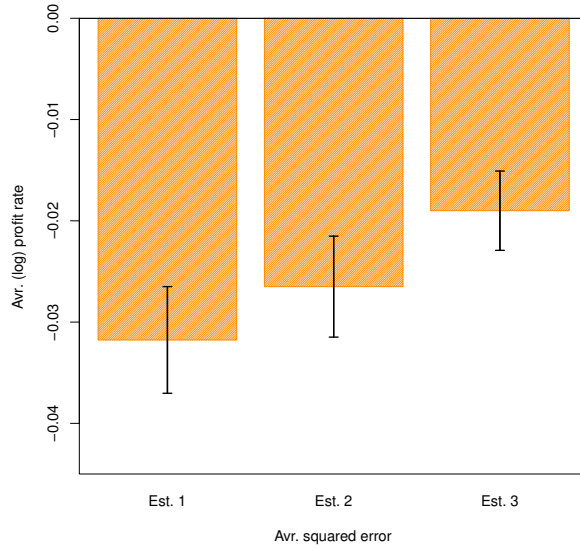
Figure 3: Outcomes and Accuracy



Panel a: Misallocation Measures



Panel b: TFP/accuracy Relationship



Panel c: Profit rate/accuracy Relationship

Note: Panel a plots the average cross-sectional dispersion (standard deviation) in $\log(\text{MPRL})$, $\log(\text{MPRK})$, and $\log(\text{TFPR})$ for the I/B/E/S-Compustat sample. Sectors are defined by their four-digit NAICS industry classification, and are weighted together by their average share in overall employment. We define a “Good” forecasting firm as one that (i) is below the median in terms of the mean-squared-error of log-revenue forecasts; and (ii) one for which we have at least five observations. “Bad” forecasting firms constitute the rest of the sample. Panel b plots the coefficient from a regression of average (log-)total factor productivity for an individual firm on its mean-squared-error of one-year-ahead (log-)revenue forecasts (Est. 1). The second column controls for firm size, age, and sector fixed effects (Est. 2). The third column further trims TFP outlier observations at the 1 percent level (Est. 3). Firm-level TFP is computed as in [Ottonello and Winberry \(2020\)](#). Panel c plots the results from analogous regressions of the average (log-)profit rate for an individual firm on its mean-squared-error of one-year-ahead (log-)revenue forecasts and controls. Sample: 2002-2022.

else equal, firms that are more accurate forecasters have less cross-sectional dispersion in marginal revenue products and TFPR. This is consistent with these firms being better able to anticipate the future demand for their products. Thus, firms that are, all else equal, better at predicting future revenue appear to misallocate resources less—in that sense that these firms have marginal revenue products that closer to each other.

This striking difference (of around 30-40 percent in measures of cross-sectional dispersion) is robust to alternative definitions of “accurate and inaccurate firms” (e.g., focusing on firms in the top quartile of the error distribution instead), sectoral definitions (e.g., considering six-digit NAICS industries), as well as a consistent feature across time. Clearly, differences in distortions unrelated to forecast accuracy may account for some of the evidence in Panel a. Nevertheless, on balance, the evidence Panel a in Figure 3 points towards systematic improvements in resource allocation as the accuracy of expectations improves.

Misallocation of scarce resources within a sector is one mechanism by which errors in expectations decrease overall productivity (David *et al.*, 2016). Panel b in Figure 3 provides additional evidence on the productivity benefits associated with more accurate firms. The panel shows the relationship between average firm-level TFP and the average accuracy of one-year-ahead revenue expectations in the cross-section, controlling for firm size, age, and sector fixed effects. Unlike Panel a, which studied the misallocation of resources *within a sector*, this Panel instead considers the *firm-level* relationship between accuracy and productivity.

Across the different regression specifications in Panel b, more accurate firms showcase higher levels of TFP. All else equal, more accurate firms are systematically more productive, even when accounting for their age, size, and other control variables. The estimated relationship, furthermore, implies that moving a firm from the 75 percentile of the squared-error of distribution to the 25 percentile is associated with, all else equal, increase in firm-level TFP by 4 percent (Column 2 Panel b). More accurate firms make more productive choices than their less informed counterparts.

Consistent with this increase in productivity, the final panel in Figure 3 (Panel c) documents that more accurate firms are also more profitable. We measure firm profitability by the logarithm of the profit rate, which is defined as $1 + \frac{\text{profits}}{\text{total costs}}$. In line with our previous evidence, across the different regression specifications, we consistently estimate firm profitability to be increasing in the accuracy of one-year-ahead forecasts—even when controlling for a firm age, size, and sector.

In sum, in this subsection, we have documented that more accurate firms in the cross-section are less misallocated, more productive, and more profitable. The Appendix (Table A.10 and A.11) further shows that more accurate firms grow faster, and that more optimistic firms also invest and employ more. Combined with the evidence in the previous subsection,

our results discipline the potential channels by which improvements in firms' accuracy have macroeconomic consequences. In the next section, we develop a heterogeneous-firm model, augmented with the choice to produce information, consistent with this evidence. We then explore the extent to which improvements in firms' accuracy and the associated increase in firm size have contributed to macroeconomic performance over the past two decades.

3 Baseline Setup

We begin by developing our theoretical framework. The next section explores the extent to which the equilibrium of our model is consistent with the empirical evidence, as well as the important role that general equilibrium interactions play for this result to be the case.

3.1 Economic Environment

Preferences and endowments. We consider an economy populated by a representative household whose objective is to maximize consumption utility net of the dis-utility of labor:

$$\mathcal{U} = C - v(L), \quad (1)$$

where $v(\cdot)$ is increasing and convex with $v(0) = v'(0) = 0$. Consumption itself is an aggregate of mass one of differentiated varieties:

$$C = \left(\int_0^1 c_i^{\frac{\theta-1}{\theta}} \cdot di \right)^{\frac{\theta}{\theta-1}}, \quad (2)$$

where c_i is the consumption of variety $i \in [0, 1]$, and $\theta > 1$ is the elasticity of substitution among varieties. Labor is used in the production both of goods and of information.

Production of goods. Each variety i is produced by a monopolistically competitive firm i , owned by the household, in accordance with the production technology:

$$y_i = A_i \cdot l_i, \quad (3)$$

where l_i are the units of labor employed by the firm, and A_i is the firm's productivity. We assume firm productivity is stochastic and distributed according to:

$$a_i \equiv \log(A_i) = \mu_i + \varepsilon_i^a, \quad \varepsilon_i^a \sim \mathcal{N}(0, \tau_a^{-1}) \quad (4)$$

where $\mu_i \in \mathbb{R}$ is the mean log-productivity of firm i , ε_i^a is the shock to firm productivity that is drawn independently of the shocks to other firms' productivities, and $\tau_a > 0$. Firms in the

economy are *ex-ante heterogeneous*, as the mean of their log-productivities, μ_i , is distributed in the population of firms according to:

$$\mu_i \sim \mathcal{N}(\mu, \tau_\mu^{-1}), \quad (5)$$

where $\mu \in \mathbb{R}$ and $\tau_\mu > 0$. Since μ_i is the only source of ex-ante heterogeneity among firms, we throughout refer to μ_i as *firm i 's type*.

Uncertainty and information production. The central friction in the economy is that firms are uncertain about their productivity when deciding how much labor to employ. In particular, when choosing l_i , the firm knows its mean log-productivity, μ_i , but it has yet to learn the realization of the shock ε_i^a . To guide its decision, however, the firm can access a signal, s_i , about its productivity with the property that:

$$s_i = a_i + \varepsilon_i^s, \quad \varepsilon_i^s \sim \mathcal{N}(0, \tau_i^{-1}), \quad (6)$$

where ε_i^s is the error of the signal, which is drawn independently of the errors of the other firms' signals, and where $\tau_i \in \{\underline{\tau}, \bar{\tau}\}$ is the signal precision. Whereas the firm can access a signal with precision $\underline{\tau} \geq 0$ at no cost, it must allocate $\chi > 0$ units of labor to information production in order to increase the precision to $\bar{\tau} > \underline{\tau}$. We denote firm i 's information production choice by $\iota_i \in \{0, 1\}$, where $\iota_i = 1$ whenever the firm produces information.

3.2 Markets and Timing

Markets. The market for labor is perfectly competitive: the household supplies labor and each firm hires the units of labor it desires to employ, taking the market wage, w , as given. In the market for variety i , the household takes the price of the variety, p_i , as given and chooses how many units of the variety to consume. By contrast, when choosing its production, the monopolistically competitive firm internalizes that its output affects the market-clearing price.

Timing. The timing of events is as follows: at the start of time, nature draws $\{\mu_i, \varepsilon_i^a, \varepsilon_i^s\}_i$, and each firm i learns its type, μ_i . The economy then proceeds through three stages. In the first stage, each firm i chooses whether or not to produce information, ι_i . In the second stage, each firm i observes its signal s_i and decides the amount of labor to employ, l_i . In the third and final stage, each firm i learns its productivity a_i , and produces with the labor that it had decided to employ. The representative household decides its demand for each variety and supplies labor to firms. Markets clear and consumption takes place.

3.3 Optimization and Equilibrium

Household problem. The representative household chooses consumption $\{c_i\}_i$ and labor L to maximize its utility in (1) and (2), subject to the budget constraint:

$$\int_0^1 p_i \cdot c_i \cdot di = w \cdot L + \int_0^1 \pi_i \cdot di, \quad (7)$$

where p_i is the price of variety i , w is the wage, and π_i are the profits of the firm producing variety i . The household takes prices, $\{p_i\}_i$, and the wage rate, w , as given, when solving its problem. The solution to the household's problem yields the demand for each variety:

$$c_i = p_i^{-\theta} \cdot C, \quad (8)$$

where we have normalized the ideal price index to one, i.e., $\int_0^1 p_i^{1-\theta} di = 1$, in addition to the labor supply schedule:

$$v'(L) = w. \quad (9)$$

Firm problem. The ex-post profits of firm i are given by the firm's revenue net of its expenditures on labor and on information:

$$\pi_i = p_i \cdot y_i - w \cdot l_i - w \cdot \chi \cdot \iota_i. \quad (10)$$

The problem of the firm can be conveniently expressed in two stages.

In the *second stage*, conditional on its information set (μ_i, s_i, τ_i) , the firm chooses labor, l_i , to maximize its expected profits from goods production (i.e., excluding information costs):

$$\tilde{\pi}_i(\mu_i, s_i, \tau_i) \equiv \max_{l_i} \mathbb{E}[p_i \cdot y_i - w \cdot l_i | \mu_i, s_i, \tau_i], \quad (11)$$

subject to feasibility (3) and the output of the variety being equal its demand (8). Here, $\mathbb{E}[\cdot | \mu_i, s_i, \tau_i]$ denotes the expectations operator conditional on (μ_i, s_i, τ_i) . When choosing l_i , the firm optimally equates the expected marginal revenue product of labor to the wage:

$$\frac{\theta - 1}{\theta} \cdot \frac{\mathbb{E}[p_i \cdot y_i | \mu_i, s_i, \tau_i]}{l_i} = w. \quad (12)$$

In the *first stage*, by contrast, conditional on μ_i , the firm chooses information production,

ι_i , to maximize its expected profits. As a result, it optimally sets:

$$\iota_i = \begin{cases} 1 & \text{if } \mathbb{E}[\hat{\pi}_i(\mu_i, s_i, \tau_i) | \mu_i, \tau_i = \bar{\tau}] - w \cdot \chi \geq \mathbb{E}[\hat{\pi}_i(\mu_i, s_i, \tau_i) | \mu_i, \tau_i = \underline{\tau}] \\ 0 & \text{otherwise} \end{cases}, \quad (13)$$

where $\mathbb{E}[\cdot | \mu_i, \tau_i]$ denotes the expectations operator conditional on μ_i and τ_i .

We are now ready to define an equilibrium of our economy.

Definition 1. *An equilibrium consists of allocations $\{c_i, y_i, l_i, \iota_i\}_i$, prices $\{p_i\}_i$ and wage w , such that (i) given the prices and the wage, the allocations solve the household and the firm problems, i.e., equations (2)-(13) hold; and (ii) the prices and the wage clear the goods and the labor markets, i.e., $c_i = y_i$ for all i and $\int_0^1 (l_i + \chi \cdot \iota_i) \cdot di = L$.*

4 Equilibrium Characterization

We proceed by studying cross-sectional outcomes at the firm level, taking economy-wide variables as given. We then turn to the aggregate implications of firms' information choices. Throughout, we discuss the extent to which our model predictions align with the empirical evidence documented in Section 2. We end the section by analyzing the aggregate effects of a decline in information costs.

4.1 Information in the Cross-section

Since the benefits to a firm of employing labor or acquiring information will depend on the level of aggregate consumption and employment in the economy, it will be useful for us to first define a notion of *market size* faced by the firms.

Definition 2. *Let $\Omega \equiv C^{\frac{1}{\theta}} \cdot v'(L)^{-1}$, which we refer to as the **market size** faced by firms.*

Thus, market size, Ω , is, all else equal, large when aggregate consumption, C , is large or when the labor costs, $w = v'(L)$, are low. As we show next, Ω is an important determinant of firm-level employment and information production.

Proposition 1. *For $\tau \geq 0$, let $g(\tau) \equiv \exp^{\frac{1}{2} \cdot (\frac{\theta-1}{\theta})^2 \cdot \frac{1}{\tau_a} \cdot \frac{\tau_a + \tau \cdot \theta}{\tau_a + \tau}}$. In equilibrium:*

(i) *Firm i of type μ_i which observes signal s_i with precision τ_i , chooses:*

$$l_i = \left(\frac{\theta - 1}{\theta} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right] \cdot \Omega \right)^{\theta}. \quad (14)$$

(ii) Firm i of type μ_i chooses:

$$l_i = \begin{cases} 1 & \text{if } \frac{1}{\theta-1} \cdot (\mathbb{E}[l_i|\mu_i, \tau_i = \bar{\tau}] - \mathbb{E}[l_i|\mu_i, \tau_i = \underline{\tau}]) \geq \chi \\ 0 & \text{if } \frac{1}{\theta-1} \cdot (\mathbb{E}[l_i|\mu_i, \tau_i = \bar{\tau}] - \mathbb{E}[l_i|\mu_i, \tau_i = \underline{\tau}]) < \chi \end{cases}, \quad (15)$$

where:

$$\mathbb{E}[l_i|\mu_i, \tau_i] = \left(\exp^{\frac{\theta-1}{\theta} \cdot \mu_i} \cdot g(\tau_i) \cdot \frac{\theta-1}{\theta} \cdot \Omega \right)^\theta. \quad (16)$$

Proposition 1 has several important implications.

Information and allocative efficiency. Part (i) of Proposition 1 reveals that a central role of information in our economy is that it helps a firm better correlate their input choices with realized productivity. All else equal, this boosts the allocative efficiency within the firm. Let tfpr_i denote the (log-)total revenue productivity of firm i :

$$\text{tfpr}_i \equiv \log(p_i \cdot y_i) - \log(l_i). \quad (17)$$

Absent information frictions, we would have that $\text{tfpr}_i = \text{tfpr}_j$ for all i, j (equation 12). Any dispersion in tfpr_i is therefore a tell-tale sign of inefficiency (relative to first-best) resulting from information frictions. Indeed, for our economy, we find that:

Corollary 1. *In equilibrium:*

$$\text{VAR}[\text{tfpr}_i|\mu_i, \tau_i] = \left(\frac{\theta-1}{\theta} \right)^2 \cdot \frac{1}{\tau_a + \tau_i} = \text{VAR}[\text{error}_i|\mu_i, \tau_i], \quad (18)$$

where $\text{error}_i \equiv \log(p_i \cdot y_i) - \log(\mathbb{E}[p_i \cdot y_i|\mu_i, \tau_i])$.

Corollary 1 establishes that input allocation is more efficient among firms with more precise information, i.e., a larger τ_i , and that, all else equal, firms with more precise information are also those that have more accurate revenue forecast errors. We will later utilize the mapping in (18) in our quantification exercise in Section 7, to help pin-down the implied decline in tfpr_i dispersion from the observed decline in uncertainty.

Information, profitability, and firm size. Part (ii) of Proposition 1 implies that information boots profitability and firm size. Notice that, from condition (12), a firm's expected revenues and profits from production are proportional to its expected employment:

$$\mathbb{E}[p_i \cdot y_i|\mu_i, \tau_i] = \theta \cdot \mathbb{E}[\hat{\pi}_i|\mu_i, \tau_i] = \frac{\theta}{\theta-1} \cdot w \cdot \mathbb{E}[l_i|\mu_i, \tau_i], \quad (19)$$

where $\mathbb{E}[l_i|\mu_i, \tau_i]$ is not only increasing in firm i 's mean productivity, μ_i , but also in the precision of its information, τ_i (see equation 16). Intuitively, because a more informed firm is able to allocate its inputs more efficiently, such a firm can earn higher revenues and profits from each unit of labor it employs, on average. Optimal input choice then requires that the firm expand its scale to equalize its marginal revenue product of labor to the wage.

Information and amplification. Part (ii) of Proposition 1 further shows that our economy features increasing returns to information production: namely, whereas the benefits to information scale with firm size—which grows with μ_i —information costs do not. Thus:

Corollary 2. *Given market size Ω , firm i produces information if and only if its type μ_i is above the marginal type:*

$$\bar{\mu}(\Omega) \equiv \frac{1}{\theta - 1} \cdot \log \left(\frac{(\theta - 1) \cdot \chi}{\left(g(\bar{\tau})^\theta - g(\underline{\tau})^\theta\right) \cdot \left(\frac{\theta-1}{\theta} \cdot \Omega\right)^\theta} \right). \quad (20)$$

Corollary 2 implies that, in the cross-section, the presence of information choice amplifies the effect of heterogeneity in productivity on firm size. Because expected firm size, $\mathbb{E}[l_i|\mu_i, \tau_i]$, is (i) increasing both in mean-productivity, μ_i , and the accuracy of information, τ_i (equation 16); and (ii) because τ_i is increasing in μ_i (equation 20), an ex-ante larger firm will, on average, produce better information, increasing its size further. As a result, the presence of information choice increases differences in the firm-size distribution. Finally, the corollary shows that a firm's incentives to produce information are increasing in market size, Ω . But, as we will see shortly, Ω may itself increase with information production, due to general equilibrium force, giving rise to strategic complementarities in information choice and to amplification at the aggregate level. Before we turn to this issue, however, we briefly discuss the extent to which our results are consistent with the empirical evidence documented in Section 2.

Model mechanisms and evidence. The cross-sectional predictions of our baseline framework are, on balance, in line with the empirical evidence. First, as discussed directly above, it is, all else equal, larger, more productive firms that acquire information, improving the accuracy of their expectations. This is consistent with the evidence in Figure 2 and Figure 3 (Panel b) showing that larger firms have more accurate forecasts and are more productive. Notice that, in our model, the tripartite relationship between firm size, accuracy of expectations, and productivity arises in part due to *selection*—it is larger, more productive firms that produce information, improving their accuracy. Section 6 provides a *causal* mechanism for this relationship, which arises due to informed firms capacity to better manage risky projects.

Second, our results are, moreover, in line with the evidence in Figure 3 (Panel a and). The main role of information in our framework is that it helps firms better their allocate inputs.

As a result, larger, more accurate firms exhibit less "misallocation of resources" (Corollary 1). They are also, as shown more profitable, as a informed a firm can earn higher revenues and profits from each unit of labor it employs, on average, which further increases its size.

Finally, we have shown that the cross-sectional predictions of our baseline framework are consistent with the empirical evidence documented in Section 2. In Section 7, we show that our model can also speak to the quantitative relevance of the effects highlighted in Section 2.

4.2 Information in General Equilibrium

We next turn to the general equilibrium of our economy. This will also help expose the forces through-which a decline in information costs affects the economy. To analyze the aggregate consequences of firms' information choices, it will be useful to first define an appropriate notion of *aggregate total factor productivity* (TFP). This notion captures the units of consumption goods that our economy can generate out of the workers allocated to production.

Definition 3. *Given aggregate consumption, C , and aggregate employment in goods production, $\mathcal{L} \equiv \int_0^1 l_i \cdot di$, we define aggregate total factor productivity, $\mathcal{A}$, as $\mathcal{A} \equiv C \cdot \mathcal{L}^{-1}$.*

Using the optimal labor choice in Proposition 1, we can express TFP as a function of firms' information sets, $\{(\mu_i, s_i, \tau_i)\}$, and hence their information choices:

$$\mathcal{A} = \left(\int_0^1 A_i^{\frac{\theta-1}{\theta}} \cdot \left(\frac{l_i}{\mathcal{L}} \right)^{\frac{\theta-1}{\theta}} \cdot di \right)^{\frac{\theta}{\theta-1}} = \left(\int_0^1 \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\theta \cdot di \right)^{\frac{\theta}{\theta-1}}, \quad (21)$$

where, to obtain the last equality, we also make use of the fact that:⁴

$$\mathcal{L} = \int_0^1 E \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\theta \cdot di \cdot \left(\alpha \cdot \frac{\theta-1}{\theta} \cdot \Omega \right)^\theta. \quad (22)$$

Recall that Corollary 2 shows that firms' information choices are fully characterized by the marginal type, $\bar{\mu}$, that is just indifferent to producing information in equilibrium. Hence:

⁴To arrive at the equality, notice that by the law of iterated expectations:

$$\begin{aligned} \int_0^1 A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\theta-1} \cdot di &= \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\theta-1} \right] \\ &= \mathbb{E}_{\mu_i, s_i, \tau_i} \left\{ \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\theta-1} | \mu_i, s_i, \tau_i \right] \right\} \\ &= \mathbb{E}_{\mu_i, s_i, \tau_i} \left[\mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\theta \right] = \int_0^1 \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\theta \cdot di. \end{aligned}$$

Lemma 1. *Given the marginal-type firm, $\bar{\mu}$, that is indifferent to producing information in equilibrium, the aggregate total factor productivity is:*

$$\mathcal{A}(\bar{\mu}) \equiv \left[\mathbf{A}(\underline{\tau})^{\theta-1} \cdot \xi(\bar{\mu}) + \mathbf{A}(\bar{\tau})^{\theta-1} \cdot (1 - \xi(\bar{\mu})) \right]^{\frac{1}{\theta-1}}, \quad (23)$$

where $\mathbf{A}(\tau) \equiv \exp^{\mu + \frac{1}{2} \cdot (\theta-1) \cdot \frac{1}{\tau\mu}} \cdot g(\tau)^{\frac{\theta}{\theta-1}}$, the weight $\xi(\bar{\mu}) \equiv \Phi\left((\mu - \bar{\mu}) \cdot \sqrt{\tau_{\mu}} - (\theta - 1) \cdot \frac{1}{\sqrt{\tau_{\mu}}}\right)$, and $\Phi(\cdot)$ is the standard normal cumulative density function.

The main takeaway from Lemma 1 is that information production, by boosting the efficiency of input allocation within each firm (Corollary 1), also raises the aggregate productivity of the economy as a whole. In one extreme, aggregate TFP equals to $\mathbf{A}(\underline{\tau})$ when (almost) no firm produces information. In another extreme, it equals to $\mathbf{A}(\bar{\tau}) > \mathbf{A}(\underline{\tau})$ when (almost) all firms chose to produce information. More generally, however, the economy's productivity falls somewhere in between these two extremes, increasing continuously as a larger share of firms begin to produce information (i.e., as $\bar{\mu}$ decreases).

Equilibrium determination. Aggregate productivity is an important determinant of economy-wide labor use, L , and consumption, C . Consider, to start, aggregate employment:

$$L = \Phi\left(-(\mu - \bar{\mu}) \cdot \sqrt{\tau_{\mu}}\right) \cdot \chi + \mathcal{A}(\bar{\mu})^{\frac{\theta-1}{\theta}} \cdot \left(\frac{\theta-1}{\theta} \cdot \Omega\right)^{\theta}. \quad (24)$$

The first term in equation (24) is the aggregate information cost, which is paid by all the firms above the marginal-type firm, $\mu_i \geq \bar{\mu}$. The second term is the aggregate employment in goods production, $\mathcal{L}$, and follows by combining equations (21) and (22). Equation (24) thus expresses the aggregate employment, L , as a function of the marginal type, $\bar{\mu}$, and the market size, Ω , both of which are endogenous of course. We can analogously express aggregate consumption as a function of $\bar{\mu}$ and Ω :

$$C = \mathcal{A}(\bar{\mu})^{1 + \frac{\theta-1}{\theta}} \cdot \left(\frac{\theta-1}{\theta} \cdot \Omega\right)^{\theta}, \quad (25)$$

where we have used the definition of aggregate TFP in addition to equation (22).

We next combine the above expressions for consumption and labor with the definition of market size to yield the equilibrium market size as a function of the marginal type $\bar{\mu}$.

Lemma 2. *Given the marginal-type firm, $\bar{\mu}$, that is indifferent to producing information in*

equilibrium, the market size that is consistent with market clearing is:

$$\Omega(\bar{\mu}) = \frac{\left[v'^{-1} \left(\frac{\theta-1}{\theta} \cdot \mathcal{A}(\bar{\mu})^{(1+\frac{\theta-1}{\theta}) \cdot \frac{1}{\theta}} \right) - \Phi \left(-(\mu - \bar{\mu}) \cdot \sqrt{\tau_{\mu}} \right) \cdot \chi \right]^{\frac{1}{\theta}}}{\frac{\theta-1}{\theta} \cdot \mathcal{A}(\bar{\mu})^{\frac{\theta-1}{\theta} \cdot \frac{1}{\theta}}}, \quad (26)$$

where $\mathcal{A}(\cdot)$ is given by Lemma 1 and $\Phi(\cdot)$ is the standard normal cdf.

Equation (26) defines a continuous schedule $\Omega : \mathbb{R} \rightarrow \mathbb{R}^+$, which for a given marginal type, $\bar{\mu}$, yields a market size, $\Omega(\bar{\mu})$ that is consistent with goods and labor market clearing. By contrast, equation (20) in Corollary 2 defines a continuous schedule $\bar{\mu} : \mathbb{R}^+ \rightarrow \mathbb{R}$ that for a given equilibrium market size, Ω , yields marginal-type firm, $\bar{\mu}(\Omega)$ that is just indifferent to producing information. Equilibria are then given by all the fixed points of the composite map $\Omega \circ \bar{\mu} : \mathbb{R}^+ \rightarrow \mathbb{R}$, i.e., μ^* satisfying:

$$\bar{\mu}(\Omega(\mu^*)) = \mu^*, \quad (27)$$

which can be shown to always exist.

Proposition 2. *An equilibrium exists. In it, an equilibrium marginal-type firm, μ^* , is given by a solution to equations (20) and (26), and equilibrium total factor productivity, $\mathcal{A}^*$, employment, L^* , and consumption, C^* , are given by:*

$$\mathcal{A}^* = \mathcal{A}(\mu^*), \quad (28)$$

$$L^* = v'^{-1} \left(\frac{\theta-1}{\theta} \cdot \mathcal{A}^{*(1+\frac{\theta-1}{\theta}) \cdot \frac{1}{\theta}} \right), \quad (29)$$

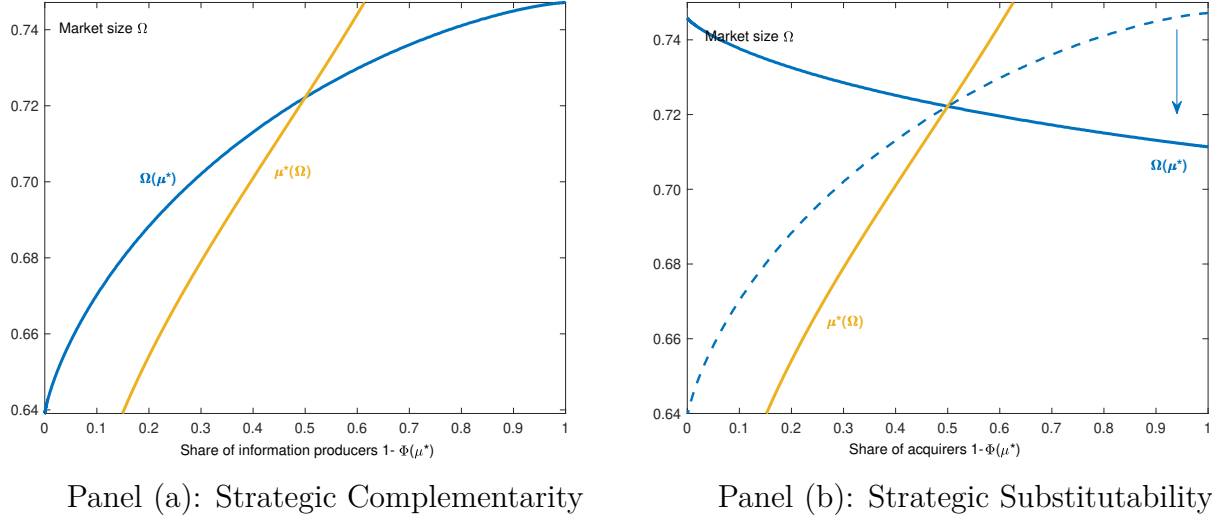
$$C^* = \mathcal{A}^* \cdot \left[L^* - \Phi \left(-(\mu - \mu^*) \cdot \sqrt{\tau_{\mu}} \right) \cdot \chi \right], \quad (30)$$

where $\mathcal{A}(\cdot)$ is given by Lemma 1 and $\Phi(\cdot)$ is the standard normal cdf.

Strategic interactions in general equilibrium. The economy-wide effects of changes in fundamentals, such as the decline in information costs that we study below, depend on whether general equilibrium forces *amplify* or *dampen* the direct effects of the change. This, in turn, will depend on whether our economy features strategic *complementarities* or *substitutabilities* in information choice. Below, we define strategic complementarities in information choice by a situation in which an increase in information production by some firms raises the market size, and thereby incentivizes other firms to produce information.

Definition 4. *There are local strategic complementarities (substitutabilities) in information choice, $\{\iota_i\}_i$, if the schedule $\Omega : \mathbb{R} \rightarrow \mathbb{R}^+$ defined by equation (26) has the property*

Figure 4: Strategic Complementarity and Substitutability



Note: Panel (a) uses the baseline calibration detailed in Section 7, whereas Panel (b) decreases the Frisch elasticity to 3. The dashed line corresponds to the schedule before the change in the Frisch elasticity.

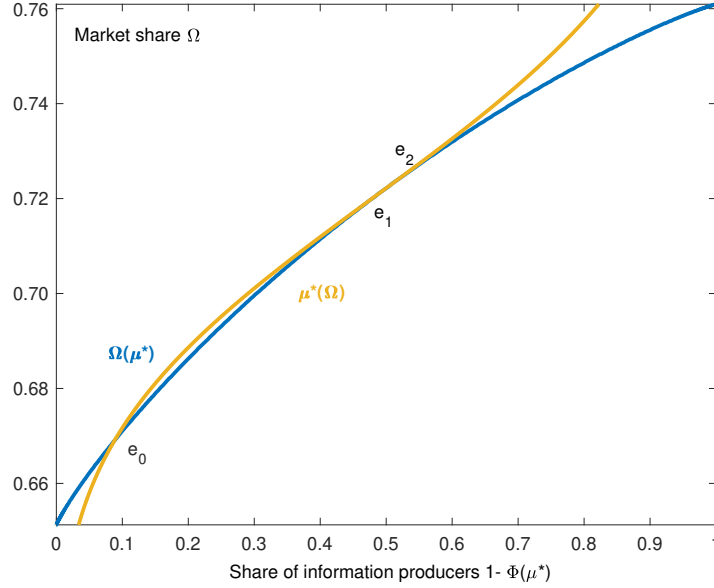
that $\Omega'(\mu^*) < 0$ ($\Omega'(\mu^*) > 0$).

Figure 4 depicts two economies that have the same equilibrium allocations, but whereas the economy in Panel (a) features strategic complementarities, the one in Panel (b) features strategic substitutabilities. Intuitively, an increase in information production (i.e., fall in $\bar{\mu}$) has two effects on the economy. On the one hand, a lower $\bar{\mu}$ triggers an increase in aggregate TFP. On the other hand, it implies an increase in labor costs, due to the larger labor demand from information producers (Section 4.1). The market size increases, and thus strategic complementarities arise, whenever the former dominates the latter. Indeed, the only difference between the two economies depicted in Figure 4 is the labor supply elasticity: Panel (a) depicts the case of a more elastic labor supply, where the rise in labor demand by the informed firms does not trigger a large increase in labor costs. This, all else equal, causes the increase in TFP to dominate the increase in labor costs. Panel (b) illustrates the opposite case. As the next corollary shows, this insight goes beyond the example depicted in Figure 4.

Corollary 3. *Consider a set of economies that have the same equilibrium allocations and are identical in all respects except for the equilibrium labor-supply elasticity.⁵ In the economy with higher labor-supply elasticity, $\Omega'(\mu^*)$ is larger, where $\Omega(\cdot)$ is defined by equation (26). Moreover, $\Omega'(\mu^*) > 0$ if the labor-supply elasticity is above a threshold.*

⁵Let L^* denote the equilibrium employment in each of these economies, indexed by j , which must coincide for all j by assumption. The equilibrium labor-supply elasticity in economy j is $\eta_j \equiv \frac{v'_j(L^*)}{v''_j(L^*) \cdot L^*}$, where $v_j(\cdot)$ is the dis-utility of labor in economy j , and where the equilibrium wage $v'_j(L^*)$ must also coincide for all j .

Figure 5: Equilibrium Multiplicity



Note: The panel uses the baseline calibration detailed in Section 7, but increases τ_μ by a factor of four.

The presence of strategic complementarities in information choice can be so strong so as to give rise to equilibrium multiplicity. To see this, consider an economy with vanishing ex-ante heterogeneity, i.e., as $\tau_\mu \rightarrow 0$. From Corollary 2, (almost) all firms choosing not to produce information is an equilibrium if and only if the information cost is above a threshold:

$$\chi > \frac{1}{\theta - 1} \cdot \exp^{(\theta-1) \cdot \mu} \cdot \left(g(\bar{\tau})^\theta - g(\underline{\tau})^\theta \right) \cdot \left(\frac{\theta - 1}{\theta} \cdot \lim_{\bar{\mu} \rightarrow -\infty} \Omega(\bar{\mu}) \right)^\theta. \quad (31)$$

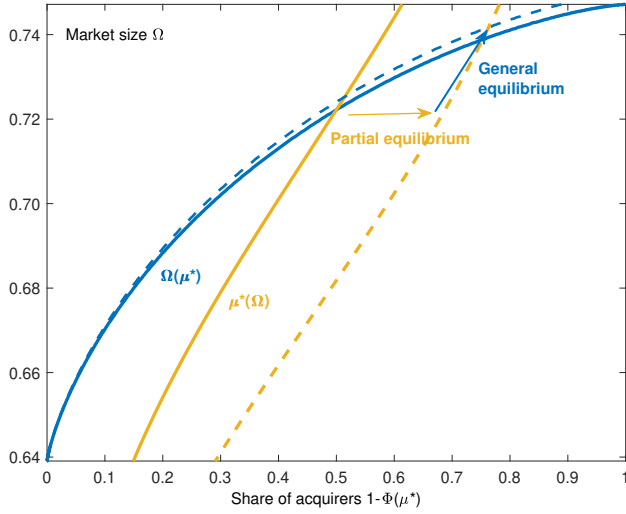
Conversely, (almost) all firms choosing to produce information is an equilibrium if and only if the information cost, χ , is below a threshold:

$$\chi \leq \frac{1}{\theta - 1} \cdot \exp^{(\theta-1) \cdot \mu} \cdot \left(g(\bar{\tau})^\theta - g(\underline{\tau})^\theta \right) \cdot \left(\frac{\theta - 1}{\theta} \cdot \lim_{\bar{\mu} \rightarrow +\infty} \Omega(\bar{\mu}) \right)^\theta \quad (32)$$

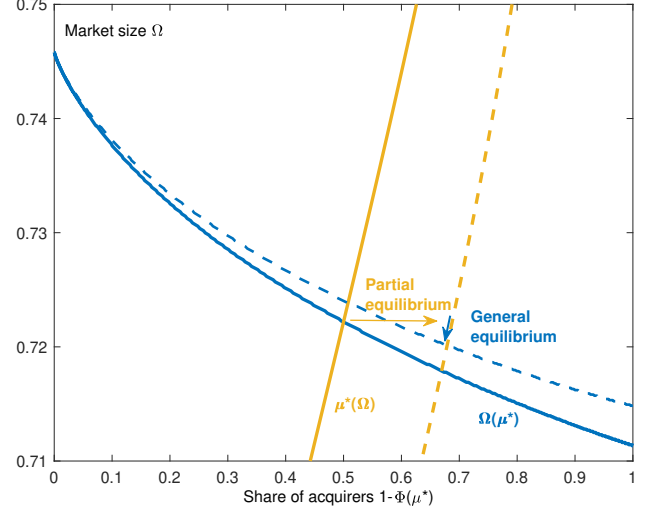
Hence, multiple equilibria arise provided that $\lim_{\bar{\mu} \rightarrow -\infty} \Omega(\bar{\mu}) > \lim_{\bar{\mu} \rightarrow +\infty} \Omega(\bar{\mu})$ and χ falls within the interval defined by the inequalities (31) and (32). Figure 5 depicts the equilibrium determination in the case in which the parameters are the same as in Figure 4, but where we reduce the amount of ex-ante heterogeneity among firms four-fold. As we can see, the forces that previously gave rise to local strategic complementarities now generate multiple equilibria.

The extent of strategic complementarities plays an important role for the aggregate effects of a decline in information cost, which as discussed in the introduction is a phenomenon that

Figure 6: A Decline in the Cost of Information χ : General Equilibrium



Panel a: Strategic Complementarity



Panel b: Strategic Substitutability

Note: Panel a uses the baseline calibration detailed in Section 7, whereas Panel b decreases the Frisch elasticity to 3. In both panels, we consider a 25 percent decline in χ from that used in the baseline calibration (Section 7). The dashed line depicts the curves after the change in the information cost.

has arguably been at play in the advanced economies over the past two decades.

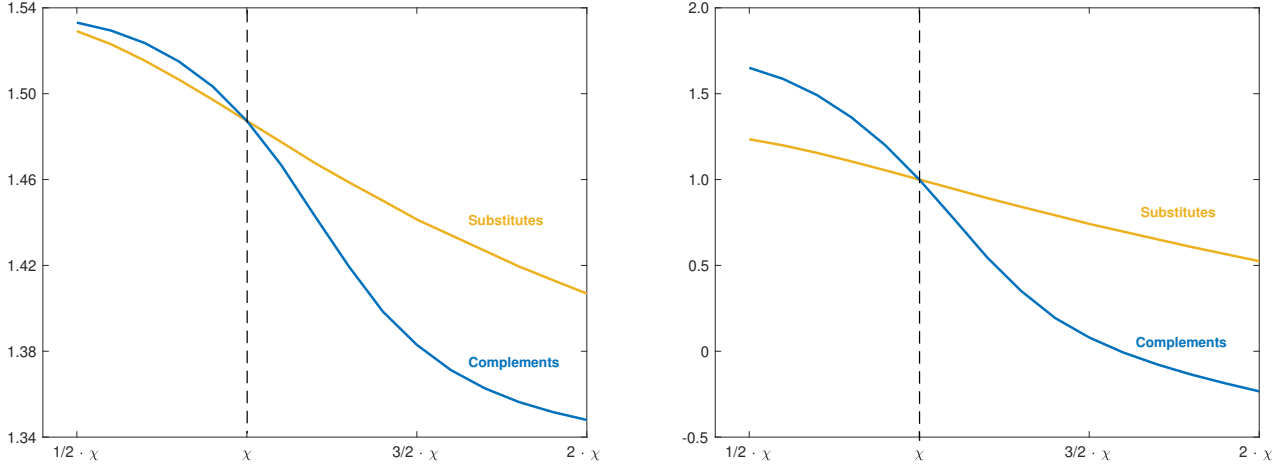
4.3 On The Effects of Declining Information Costs

In this section, we study the economy's response to a decline in information costs, which is meant to capture the broad-based improvements in data-processing technologies witnessed over the past two decades (Baley and Veldkamp, 2025). In Section 7, we explore the quantitative implications of such improvements in further detail.

We trace-out the effects of a decline in the information cost parameter, χ , on the general equilibrium of our economy in Figure 6. All else equal, a decline in information costs makes information production more attractive. In both Panels (a) and (b), this pushes out the $\bar{\mu}(\cdot)$ -curve given by equation (20), characterizing firms' incentives to produce information for a given market size, Ω . All else equal, this increases the share of information producers in the economy. Indeed, this shift corresponds to the economy's *partial equilibrium* response. The overall effect of the decline in information costs on the share of information producers, however, crucially depends on the nature of strategic interactions.

Panel (a) in Figure 6 depicts the case when information choices are *strategic complements*. As the firms' information improves, firms choose to acquire more labor inputs, which they also utilize more efficiently (Section 4.1). This, in turn, increases the overall market size of

Figure 7: A Decline in the Cost of Information χ : TFP and Welfare



Panel a: Total Factor Productivity, $\mathcal{A}$

Panel b: Welfare, $\mathcal{U}$

Note: Both panels use the baseline calibration detailed in Section 7. The figure consider changes in the cost of information χ . Panel a plots the effects on TFP, $\mathcal{A}$; Panel b depicts the effects on household welfare, $\mathcal{U}$. We scale the levels of household welfare so that they are identical when χ is at its baseline value. The case with strategic substitutes decreases the Frisch elasticity to 3.

the economy, Ω , by increasing consumption, which further boosts firms' incentives to produce information. In addition, after the decline in χ , the economy also uses fewer resources on information production itself, which causes the $\Omega(\cdot)$ -curve given in equation (26) to shift upwards, increasing market size further. Both of these general equilibrium forces *amplify* the increase in information production by firms.

By contrast, Panel b in Figure 6 depicts the case when information choices are *strategic substitutes*. The $\Omega(\cdot)$ -curve in equation (26) still shifts up, due to fewer resources being spent on information production. But the countervailing decrease in market size, Ω , here caused by higher labor costs, pushes in the opposite direction. Indeed, for the parameterization used in Panel (b), the latter effect dominates and thus the general equilibrium forces *dampen* the increase in information production by firms.

As we show next, these differences in responses caused by differences in the nature of strategic interactions, moreover, are useful to understand the aggregate effects that changes in information costs have on the economy.

Proposition 3. *Consider the most (least) informative equilibrium, i.e., equilibrium with the minimal (maximal) μ^* .⁶ Then, a local decline in the information cost, χ , increases the equilib-*

⁶The statement holds of any equilibrium in which $\Omega(\cdot)$ schedule defined by equation (26) is flatter than the inverse of the schedule $\bar{\mu}(\cdot)$ defined by equation (20) when evaluated at the equilibrium μ^* . Other equilibria (if they exist) can be shown to be unstable using a simple heuristic argument.

rium total factor productivity, $\mathcal{A}^*$, consumption, C^* , employment, L^* , and household welfare, $\mathcal{U}^*$. Moreover, the equilibrium effects of the decline in χ are larger whenever, all else equal, the local strategic complementarities are stronger.

Panels (a) and (b) in Figure 7 illustrate the effects that changes in χ have on total factor productivity, $\mathcal{A}$, and on household welfare, $\mathcal{U}$. The larger increase in the share of information producers in the case with strategic complements—that is, the larger decline in μ^* —increases total factor productivity by more in response to a decline in information costs (Panel (a)). This, in turn, reflects that a larger share of firms, in this case, make more efficient input choices. The implication of the larger increase in the efficiency by which firms make their choices is that household welfare, in this case, also increases by more (Panel (b)). The increase in the resources devoted to information production is more than compensated for by the larger increase in overall productivity, making information production invariably beneficial.

5 Is There Too Little Information?

In this section, we shift focus and turn to the normative properties of our economy. Our aim is to understand whether the laissez-faire equilibrium suffers from under- or over-production of information. To speak meaningfully about any discrepancy between the equilibrium and socially optimal production of information, we nevertheless first need to establish an appropriate welfare benchmark. Because firms' optimal (full-information) choices only depend on their own individual-specific productivity, we can formalize the social planner's problem as:

$$\max_{\{\iota_i(\mu_i), l_i(\mu_i, \tau_i, s_i)\}} \mathcal{U} = \underbrace{\left(\int_0^1 A_i^{\frac{\theta-1}{\theta}} \cdot l_i(\mu_i, \tau_i, s_i)^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}}}_{\equiv C} - v(L) \quad (33)$$

subject to the resource and information constraints:

$$L = \int_0^1 (l_i(\mu_i, \tau_i, s_i) + \chi \cdot \iota_i(\mu_i)) \cdot di, \quad s_i \sim \mathcal{N}(\log A_i, \underline{\tau}^{-1} + \iota_i \cdot (\bar{\tau}^{-1} - \underline{\tau}^{-1})). \quad (34)$$

In what follows, whenever needed to avoid confusion, we will use the superscripts *sp* and *de* to distinguish between the *social planner's* and the *decentralized equilibrium* allocations.

The following proposition characterizes the solution to the social planner's problem.

Proposition 4. *Let $\bar{\mu}(\cdot)$ be defined by equation (20) and Ω^{sp} be the maximal solution to:*

$$\Omega^{sp} = \frac{\left[v'^{-1} \left(\mathcal{A}(\bar{\mu}(\Omega^{sp}))^{(1+\frac{\theta-1}{\theta}) \cdot \frac{1}{\theta}} \right) - \Phi \left(-(\mu - \bar{\mu}(\Omega^{sp})) \cdot \sqrt{\tau_\mu} \right) \cdot \chi \right]^{\frac{1}{\theta}}}{\mathcal{A}(\bar{\mu}(\Omega^{sp}))^{\frac{\theta-1}{\theta} \cdot \frac{1}{\theta}}}, \quad (35)$$

where $\mathcal{A}(\cdot)$ is given by Lemma 1 and $\Phi(\cdot)$ is the standard normal cdf.

Then, at the social planner's allocation:

(i) Firm i of type μ_i which observes signal s_i with precision τ_i , chooses:

$$l_i^{sp} = \left(\mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right] \cdot \Omega^{sp} \right)^\theta. \quad (36)$$

(ii) Firm i of type μ_i chooses $l_i^{sp} = 1$ if and only if its type is above the marginal type:

$$\mu^{sp,*} = \bar{\mu}(\Omega^{sp}). \quad (37)$$

Analogous to the decentralized equilibrium allocations, the presence of strategic complementarities can lead to multiple solutions to the first-order conditions that characterize the social planner's solution. The efficient outcome maximizes welfare within this set of outcomes.

The first part of the proposition characterizes firms' efficient labor choices. It shows that, for a given market size, Ω , the social planner's solution features more employment by all firms. Indeed, for a given Ω , the only difference between the decentralized level of employment in equation (14) and the socially optimal one in equation (36) is the distortion to due markup, $\frac{\theta}{\theta-1}$. Thus, for a given market size, the presence of monopolistic competition depresses firm employment and output (Blanchard and Kiyotaki, 1987).

The second part of the proposition, in contrast, focuses on firms' information choices. As in the linear-quadratic beauty-contest games studied in Angeletos and Pavan (2007) and Myatt and Wallace (2011), efficiency of actions here also implies efficiency in the use of information. Conditional on an efficient market size—that is, when $\Omega^{de} = \Omega^{sp}$ —the equilibrium cut-off for when to produce information in (20) coincides with the socially optimal cut-off in (??). When actions, and hence the market size, are efficient so too are firms' information production choices. In this case, private incentives coincide with the social planner's.

The final part of the proposition combines the earlier insights. It shows that, because of the inverse markup associated with monopolistic competition, $\frac{\theta-1}{\theta}$, the socially optimal market size exceeds that in the decentralized equilibrium, $\Omega^{sp} > \Omega^{de}$. The depressed level of employment that arises in the decentralized equilibrium causes too few goods to be produced, and, hence, a too small market size. Consequently, too little information is produced in equilibrium.

The flip-side of the conditional efficiency of information production is, however, that a simple input subsidy can restore efficiency to both firms' actions and firms' information production choices, as shown in the following corollary.

Corollary 4. *The social planner's allocation can be decentralized with a subsidy, ω to firms*

for each unit of labor employed, whether in goods or information production, given by:

$$\omega = 1 - \frac{\theta}{\theta - 1} \cdot \frac{\Omega(\omega)}{\Omega^{sp}}, \quad (38)$$

where $\Omega(\omega) \equiv \frac{C^*(\omega)^{\frac{1}{\theta}}}{v'(L^*(\omega))}$ denotes the market size with the subsidy, and $C^*(\omega)$ and $L^*(\omega)$ are the equilibrium consumption and employment in the economy with the subsidy ω .

At first pass, there are two separate distortions to the equilibrium of our economy: (i) market power, caused by the presence of imperfect (monopolistic) competition, which distorts labor choices downward (part (i) of Proposition 4); and (ii) coordination failures, which can arise due to increasing returns to information production. The former implies that a simple, appropriately chosen subsidy on inputs can render the laissez-faire equilibrium efficient. However, if strategic complementarities in information choice are strong enough to generate multiple equilibria, the input subsidy must also depend on the equilibrium market size in the economy. Note that, although under the optimal subsidy in Corollary 4 the equilibrium subsidy will still equal $1 - \frac{\theta}{\theta-1}$; however, off-equilibrium, the subsidy would be larger if the firms were to attempt and coordinate on less informative equilibria, rendering such equilibria inconsistent with optimization and market clearing.

6 Information and Technology Choice

To be completed.

7 A Quantitative Exploration

To be completed.

8 Final Remarks

The advance of data-processing technologies has the potential to transform most aspects of economic life. In this paper, we have focused on a particular facet of the ongoing data revolution: its capacity to systematically improve firms' economic forecasts. Using micro data on managerial forecasts, we have documented a systematic increase in the forecasting ability of US firms over the past two decades, and shown that this increase is tightly tied to changes in the firm-size distribution. To explore the macroeconomic consequences of these developments, we have proposed a quantitative-theoretical framework that is consistent with the evidence.

Our framework demonstrates that firms' size-dependent incentives to use information-driven decision-making can rationalize the size-forecasting-ability relationship observed in the

survey data. Consistent with our estimates, our model shows that firms that use information more intensely allocate inputs more efficiently, adopt more efficient technologies, are more profitable, and grow faster and larger. We have illustrated the macroeconomic consequences of these cross-sectional shifts in firm performance. For plausible parameter values, consistent with the observed increase in forecast performance, we have documented that total factor productivity (household welfare) in the US would have been 10 percent (8 percent) lower in 2022 absent the increase in the forecasting ability of US firms. We have further decomposed this overall estimate into a short-run and a long-run component. On balance, we find that that most (around 2/3) of the benefits accrue over the longer horizon, especially as firms start to adopt more efficient technologies and the increase in firm size materializes.

Finally, our results suggest that a large share of the macroeconomic consequences of information-processing arise due to general equilibrium interactions between firms. Beyond the analysis in this paper, we see important scope for extending our framework to also account for how advances in data technologies alter the way households decide which goods to purchase, and thereby their interactions with firms. Another avenue for future research is to combine models of optimal information production with those in which increases in firm size have detrimental welfare consequences, due to decreases in firm competition ([Edmond *et al.*, 2023](#)). The latter approach would allow for a more granular welfare assessment of the recent change in the firm-size distribution—one which separates the potential costs from the potential benefits. Overall, we view the research in this paper as a useful step towards a macroeconomic framework that accounts for the ongoing changes in data-processing technologies based on a minimal set of departures from workhorse macroeconomic models.

A Motivating Evidence

A.1 Additional Estimates

Table A.2: Time Evolution of Accuracy and Size

	<i>Panel a: revenue errors and time</i>			
	Absolute error		Squared error	
	(1)	(2)	(3)	(4)
Time	−0.024*** (0.003)	−0.015*** (0.002)	−0.028*** (0.007)	−0.024*** (0.005)
Constant	1.251*** (0.037)	1.135*** (0.026)	1.303*** (0.085)	1.219*** (0.065)
Observations	12,566	12,562	12,566	12,562
Covid dummy	×	✓	×	✓
Residual std. error	1.835	1.282	4.302	3.148
F statistic	66.99***	22.21***	17.68***	22.57***
	<i>Panel b: size and time</i>			
	50th perc.	70th perc.	80th perc.	90th perc.
	(1)	(2)	(3)	(4)
Time	0.011*** (0.002)	0.012*** (0.001)	0.009*** (0.001)	0.005*** (0.001)
Constant	0.567*** (0.015)	0.374*** (0.016)	0.215*** (0.007)	0.111*** (0.005)
Observations	21	21	21	21
Residual std. error	0.029	0.032	0.021	0.014
F statistic	119.56***	116.05***	153.64***	99.41***

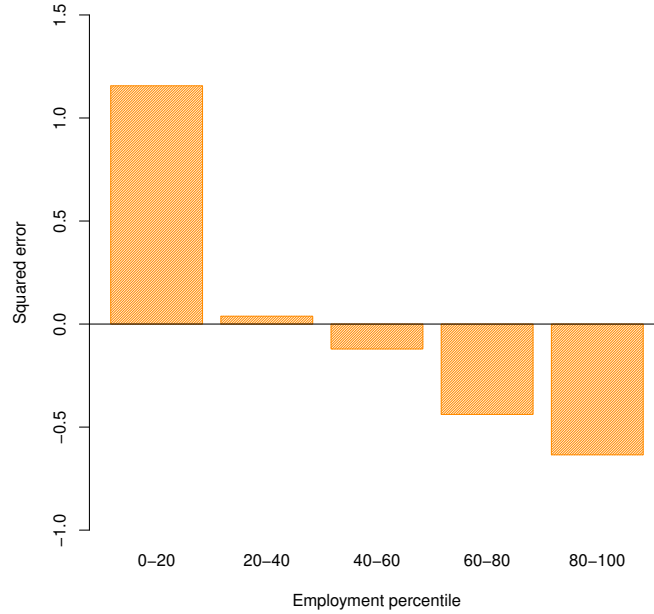
Notes: Panel least-squares estimates from the merged I/B/E/S-Compustat sample. Panel a: estimate of the coefficient of the absolute value (squared value) of individual one-year ahead revenue errors on time. Revenue errors are scaled by a firm's tangible capital stock and normalized by the overall average absolute (squared) error in the sample. The top and bottom 1 percent of errors have been removed. Panel b: estimate of the coefficient of the share of firms with employment greater than the x th percentile of firms in 2002 on time. Columns (1) and (3) are in levels, whereas Columns (2) and (4) pertain to the logs of variables. Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

Table A.3: Time Evolution of Asset Size

	<i>Panel b*: asset size and time</i>			
	50th perc.	70th perc.	80th perc.	90th perc.
	(1)	(2)	(3)	(4)
Time	0.019*** (0.001)	0.025*** (0.002)	0.027*** (0.002)	0.018*** (0.003)
Constant	0.557*** (0.014)	0.387*** (0.020)	0.232*** (0.018)	0.066*** (0.023)
Observations	21	21	21	21
Residual std. error	0.026	0.039	0.048	0.064
F Statistic	445.104***	310.202***	239.202***	62.681***

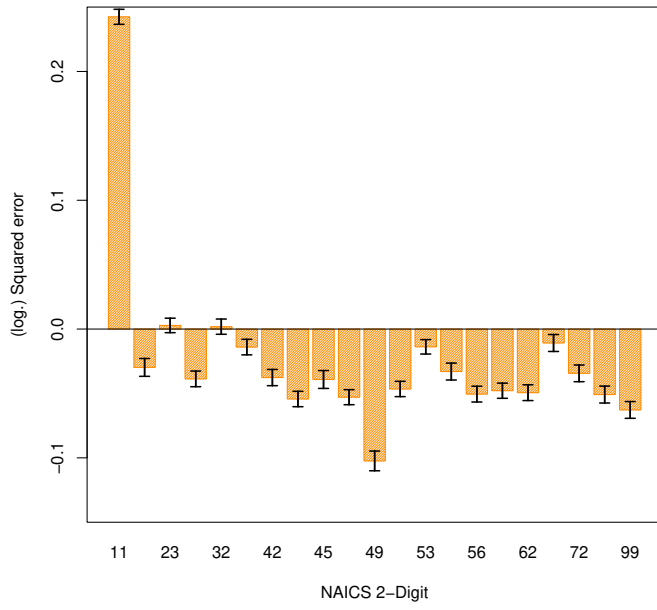
Notes: Panel least-squares estimates from the merged I/B/E/S-Compustat sample. Panel b*: estimate of the coefficient of the share of firms with assets greater than the x th percentile of firms in 2002 on time. Robust standard errors in parentheses. The 50th, 70th, 80th, and 90th percentile of the 2002-asset distribution correspond to c. 288, 512, 992, and 3,728 million USD. Sample: 2002–2022.

Figure A.8: Revenue Expectations Across the Asset Distribution

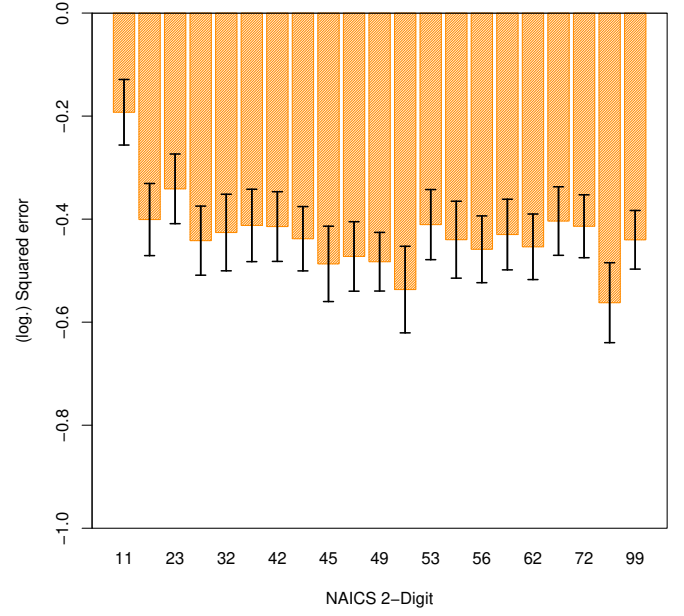


Note: The figure plots the difference between the average squared error of one-year-ahead log-revenue forecasts from the I/B/E/S-Compustat merger within size (asset) quintiles and the overall average taken across all size levels. Revenue errors are scaled by a firm's tangible capital stock and normalized by their mean value in the sample. Table A.4 reports the coefficient estimates, controlling for firm characteristics. Sample: 2002–2022.

Figure A.9: Sectoral Heterogeneity and the Time and Size Relationship



Panel a: Time Relationship



Panel b: Size Relationship

Note: Panel least-squares estimates from the merged I/B/E/S-Compustat sample. Panel a: estimate of the coefficient of the squared value of individual one-year-ahead (log-) revenue errors on time for different NAICS 2-digit sectors. Panel b: estimate of the coefficient of the squared value of individual one-year-ahead (log-) revenue errors on firm size for different NAICS 2-digit sectors. Firm size is measured by the quintile the firm's employment is at time t relative to the 2002-employment distribution. Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

Table A.4: Robustness of Revenue Expectations, Firm Size, and Time Relationship

	<i>Absolute error</i>		<i>Squared error</i>		<i>Squared error log</i>
	(1)	(2)	(3)	(4)	(5)
Firm size	−0.353*** (0.034)	−0.293*** (0.040)	−0.560*** (0.072)	−0.450*** (0.084)	
Firm assets					−0.272*** (0.056)
Time	−0.006 (0.006)		−0.003 (0.012)		
Firm age	0.004 (0.016)	0.023 (0.024)	0.046 (0.035)	0.058 (0.052)	0.022 (0.031)
Rev. volatility		0.010*** (0.002)		0.010** (0.002)	0.004 (0.011)
Observations	12,489	6,819	12,489	6,819	6,834
Time FE	×	✓	×	✓	✓
Sector FE	✓	✓	✓	✓	✓
F Statistic	10.460***	7.704***	5.494***	5.043***	2.083***

Notes: Panel estimates from the merged I/B/E/S-Compustat sample. Column (1) shows estimates from a regression of the absolute value of individual one-year-ahead revenue errors on firm size (employment), controlling for time and firm age and sector fixed effects (NAICS-4). Firm size is measured based on which quintile the firm's employment level is at time t relative to the 2002-employment distribution. Column (2) considers the same regression specification but includes time fixed effects, as well as the rolling four-year volatility of revenue. Columns (3) and (4) consider the same specifications studied in Columns (1) and (2), but instead use the squared value of individual errors as the dependent variable. Finally, Column (5) uses the squared value of individual one-year-ahead revenue errors, and measures firm size based on which quintile the firm's asset level is at time t relative to the asset distribution. Revenue errors are scaled by a firm's tangible capital stock and normalized by the overall average absolute (squared) error. The top and bottom 1 percent of forecast errors have been removed. Robust (clustered) standard errors in parentheses. Sample: 2002-2022.

Table A.5: Revenue Expectations, Firm Size, and Fixed Effects

	<i>Squared log-revenue error</i>		
	(1)	(2)	(3)
Firm size	−0.254*** (0.057)	−0.438*** (0.054)	−0.230*** (0.044)
Firm age	−0.044 (0.033)		−0.057* (0.030)
Revenue volatility	−0.014*** (0.002)		−0.014*** (0.002)
Observations	12,488	12,488	12,488
Sector FE	✓	×	×
Time FE	✓	×	×
Time×Sector FE	×	✓	✓
F statistic	4.020***	2.401***	2.530***

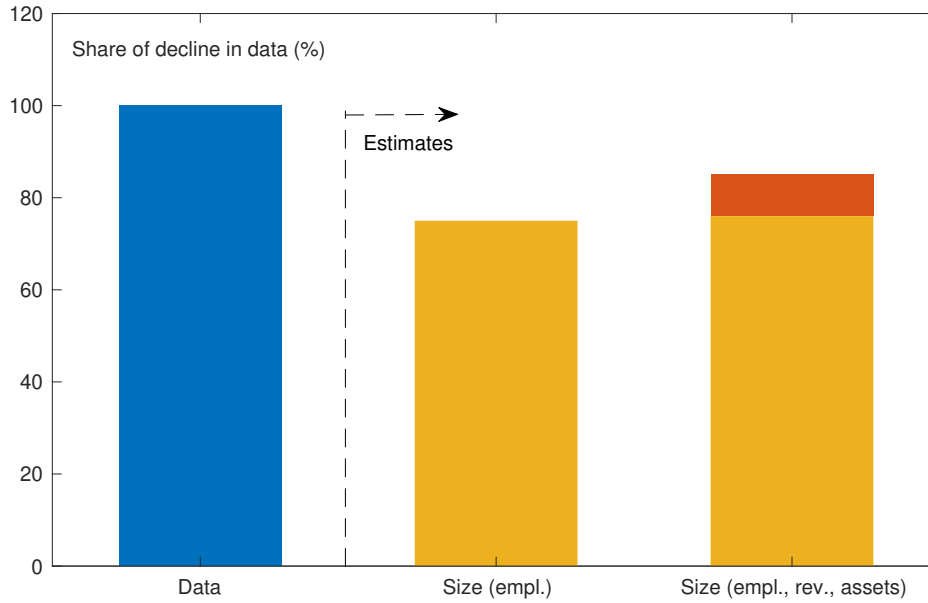
Notes: Panel estimates from the merged I/B/E/S-Compustat sample. Column (1) shows estimates of the squared value of one-year-ahead log-revenue errors on firm size (employment) and sector (NAICS-4) and time fixed effects. We also control for firm age and the individual four-year-rolling average of the volatility of revenue. Firm size is measured by the quintile the firm's employment is at time t relative to the 2002-employment distribution. Columns (2) and (3) adds time×sector (NAICS-2) fixed effects. Revenue errors are scaled by firm capital and normalized by the overall average absolute error. The top and bottom 1 percent of errors have been removed. Robust (clustered) standard errors in parentheses. Sample: 2002-2022.

Table A.6: Percentage Increase in Revenue

	<i>Absolute error</i>		<i>Squared error</i>		
	(1)	(2)	(3)	(4)	(5)
Firm size	−0.210*** (0.025)	−0.177*** (0.031)	−0.372*** (0.056)	−0.308*** (0.064)	
Time	0.0005 (0.004)		0.005 (0.008)		−0.014** (0.007)
Firm age	−0.050*** (0.016)	0.009 (0.025)	−0.095** (0.039)	0.026 (0.051)	
Rev. volatility (pct.)		0.004*** (0.001)		0.005*** (0.002)	
Observations	10,083	5,700	10,083	5,700	10,138
Time FE	×	✓	×	✓	×
Sector FE	✓	✓	✓	✓	×
F Statistic	5.589***	4.648***	3.491***	3.163***	6.747***

Notes: Panel estimates from the merged I/B/E/S-Compustat sample. Column (1) shows estimates from a regression of the absolute value of individual one-year-ahead revenue growth errors (in pct.) on firm size (employment), controlling for time and firm age and sector fixed effects (NAICS-4). Firm size is measured based on which quintile the firm's employment level is at time t relative to the 2002-employment distribution. Column (2) considers the same specification but includes time fixed effects, as well as the rolling four-year volatility of revenue. Columns (3) and (4) consider the same specifications studied in Columns (1) and (2), but instead use the squared value of individual errors as the dependent variable. Finally, Column (5) considers the raw correlation with time. Revenue errors are scaled by a firm's tangible capital stock and normalized by the overall average absolute (squared) error. The top and bottom 1 percent of forecast errors have been removed. Robust (clustered) standard errors in parentheses. Sample: 2002-2022.

Figure A.10: Size and Uncertainty Simulation



Note: The figure shows the overall decline in revenue uncertainty in the I/B/E/S-Compustat sample (Figure 1) from its average value in 2002-2005 to 2022 [data]. The figure compares the decline to that implied from the change in the firm-size distribution only, using the estimates in Table I Column 4 (with/without the inclusion of real firm assets and real firm revenue as further control variables). Firm real revenue (assets) are measured by the quintile the firm's revenue (assets) are in at time t relative to the 2002-revenue (asset) distribution. Firms revenue and assets are deflated by CPI-U.

Table A.7: Other Variables (Profits and CAPEX)

<i>Panel a: errors and time</i>				
	Profits		Capex	
	Abs. error	Sqr. error	Abs. error	Sqr. error
Time	−0.039*** (0.010)	−0.068*** (0.024)	−0.012*** (0.003)	−0.001 (0.011)
Constant	1.398*** (0.126)	1.691*** (0.303)	1.126*** (0.035)	0.986*** (0.120)
Observations	2,487	2,487	1,839	1,839
Residual std. error	2.482	6.187	1.871	6.982
F statistic	15.27***	7.385***	12.91***	0.011
<i>Panel b: errors and size</i>				
	Profits		Capex	
	Abs. error	Sqr. error	Abs. error	Sqr. error
Firm size	−0.361*** (0.082)	−0.578*** (0.195)	−0.074*** (0.018)	−0.130*** (0.047)
Firm age	−0.029 (0.080)	0.047 (0.147)	−0.070*** (0.016)	−0.117* (0.061)
Constant	0.456 (0.314)	0.680 (0.486)	−0.113*** (0.042)	−0.144 (0.165)
Observations	2,487	2,487	1,839	1,839
Sector FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
Residual std. error	2.347	6.209	1.740	6.744
F statistic	2.694***	1.056	7.906***	2.849***

Notes: Panel least-squares estimates from the merged I/B/E/S-Compustat sample. Panel a: estimate of the coefficient of the absolute value (squared value) of individual one-year ahead profit (capex) errors on time. Forecast errors are scaled by a firm's tangible capital stock and normalized by the overall average absolute (squared) error in the sample. The top and bottom 1 percent of errors have been removed. Panel b: estimate of the coefficient of the absolute value (squared value) of individual one-year ahead errors on firm size, controlling for firm age, and time and sector (NAICS level 4) fixed effects. Firm size is measured as in Table I. Robust (clustered) standard errors in parentheses. Sample: 2004–2022 (profits) and 2002–2022 (capex).

Table A.8: Output Growth Expectations from the Duke CFO Survey

	Squared error		Absolute error	
	(1)	(2)	(3)	(4)
Firm size	−0.077** (0.031)	−0.060* (0.032)	−0.050*** (0.016)	−0.045*** (0.016)
GDP familiarity		0.033 (0.061)		0.018 (0.030)
Constant	1.639*** (0.561)	1.676*** (0.650)	1.648*** (0.286)	1.681*** (0.331)
Observations	1,584	1,464	1,584	1,464
Sector FE	✓	✓	✓	✓
Time FE	✓	✓	✓	✓
R ²	0.177	0.201	0.241	0.271
Residual std. error	1.578	1.544	0.812	0.793
F Statistic	19.774***	20.165***	29.175***	29.814***

Notes: Estimates from the Duke CFO Survey. Column (1) shows estimates from a regression of the square of individual one-year-ahead real GDP growth errors on firm size (employment) and sector and time fixed effects. Firm size is measured discretely (values 1-5), depending on which quintile firm employment is in relative to the 2020-employment distribution. Column (2) controls for the familiarity of the respondent to the concept of GDP. Columns (3) and (4) consider the absolute value of individual errors. GDP errors are normalized by the overall average squared (absolute) error in the sample. The top and bottom 1 percent of errors have been removed. Robust standard errors in parentheses. Sample: 2020-2022.

Table A.9: Large Acquisitions and Forecast Accuracy

	Squared error (1)	Absolute error (2)
Large acquisitions	−0.139* (0.007)	−0.060*** (0.033)
Large acquisitions(−1)	−0.120* (0.066)	−0.083*** (0.031)
Large acquisitions(−2)	−0.106* (0.063)	−0.079*** (0.031)
Large acquisitions(−3)	0.035 (0.067)	−0.043 (0.031)
Firm age	0.037 (0.034)	0.004 (0.017)
Revenue volatility	0.002*** (0.000)	0.001*** (0.000)
Observations	5,108	5,108
Sector FE	✓	✓
Time FE	✓	✓
Residual std. error	2.325	1.050
F statistic	1.535***	2.879***

Notes: Panel least-squares estimates from the merged I/B/E/S-Compustat sample. The table estimates the coefficient of the squared value (absolute value) of individual one-year-ahead log-revenue errors on firm acquisitions, controlling for firm age, revenue volatility (rolling 4-year average), and time and sector (NAICS level 4) fixed effects. Errors are scaled by a firm's tangible capital stock and normalized by the overall average squared (absolute) error in the sample. The top and bottom 1 percent of errors have been removed. A large acquisitions is defined as one above 5 percent of a firm's assets, consistent with the definition in [Ottonello and Winberry \(2020\)](#). Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

Table A.10: LMRPK and Errors Relationship

	<i>logarithm of mrpk</i>		
	(1)	(2)	(3)
Log. revenue error firms	0.225*** (0.052)	0.371*** (0.052)	0.404*** (0.053)
Observations	12,562	12,562	12,562
Sector FE	✓	×	×
Time FE	✓	✓	✓
Sector × time FE	×	✓	×
Firm FE	×	×	✓
F statistic	35.451***	7.404***	135.318***

Notes:

Table A.11: Breakdown of the Markup/Accuracy Relationship

	<i>Avr. markup level</i>		<i>Avr. profit rate log.</i>	<i>Avr. fixed cost rate log.</i>
	(1)	(2)	(3)	(4)
Avr. squared error	-0.027 (0.034)	-0.021** (0.009)	-0.027*** (0.007)	0.014** (0.006)
Avr. firm size	-0.109* (0.066)	-0.006 (0.023)	0.054*** (0.010)	-0.095*** (0.012)
Avr. firm age	-0.102 (0.110)	-0.011 (0.015)	0.016** (0.007)	-0.022** (0.012)
Observations	2,490	2,490	2,376	2,393
Sector FE	✓	✓	✓	✓
R^2	0.108	0.320	0.311	0.534

Notes: Panel least-squares estimates from the I/B/E/S-Compustat sample. The table shows estimates of the breakdown in equation (??) of a firms (log.) markup into the profits share and the cogs share. Robust (clustered) standard errors in parentheses. Sample: 2002-2022.

Table A.12: Accuracy and Intangible Capital

	<i>Firm uncertainty</i>	
	Sqr. error	Abs. error
Firm acq. stock of intangibles	-0.070* (0.040)	-0.038*** (0.015)
Firm size	-0.405*** (0.041)	-0.211*** (0.019)
Firm age	-0.051 (0.034)	-0.034*** (0.013)
Observations	11,371	11,371
Sector FE	✓	✓
Time FE	✓	✓
F statistic	3.721***	6.499***

Notes: Panel least-squares estimates from the I/B/E/S-Compustat sample. The table estimates the relationship between of the stock of acquired intangibles and the uncertainty surrounding firms' log-revenue forecasts. The stock of acquired intangibles accounts adjusts amortization and take-outs financial goodwill (Compustat: INTAN+AM-GDWL), and the nominal stock is deflated. This is in accordance with [Chiavari and Goraya \(2023\)](#). Column (1) considers the squared value of individual errors, while Column (2) considers the absolute value. Errors are normalized by the overall average squared (absolute) error in the sample. The top and bottom 1 percent of errors have been removed. All estimates controls for time and sector (NAICS-4) fixed effects. Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

Table A.13: Optimism and Firm-level Outcomes

	<i>Realized increase (%)</i>	
	Employment	Capital
Revenue forecast (%)	0.069*** (0.026)	0.222*** (0.085)
Firm age (quintile)	-2.072 (1.627)	-5.794*** (4.198)
Observations	10,260	10,277
Firm FE	✓	✓
Time FE	✓	✓
F statistic	52.856***	137.772***

Notes: Panel least-squares estimates from the I/B/E/S-Compustat sample. The table estimates the relationship between realized growth in employment and capital and the firm-specific forecasts of revenue growth. Forecasts related to the top and bottom 1 percent of forecast errors have been removed. All estimates controls for time and firm fixed effects. Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

Table A.14: Accuracy and Productivity

	<i>Avr. (log.) total factor productivity</i>		
	(1)	(2)	(3)
Avr. squared error	-0.021*** (0.007)	-0.027*** (0.008)	-0.019** (0.008)
Avr. firm age	—	-0.008 (0.012)	-0.040 (0.043)
Avr. firm size	—	-0.012 (0.000)	-0.012 (0.000)
Observations	1,871	1,871	1,834
Sector FE	×	✓	✓
R^2	0.022	0.104	0.104

Notes: Panel least-squares estimates from the I/B/E/S-Compustat sample. The table estimates the relationship between the (cross-sectional) average squared (log-) revenue error and average (log-) total factor productivity. Firm-specific total factor productivity is estimated in accordance with the approach in [Ottonello and Winberry \(2020\)](#). All other variables are as defined in Table I. Revenue errors are normalized by the overall average squared value in the sample. The top and bottom 1 percent of errors have been removed. Column (3) further trims the top and bottom 1 percent of (log-) total factor productivity. Sector (NAICS-6) fixed effects are used. Robust (clustered) standard errors in parentheses. Sample: 2002–2022.

A.2 Data Construction: I/B/E/S-Compustat

In our analysis of firms’ forecasts, we use a combination of the I/B/E/S managerial guidance database and Compustat Fundamentals Annual. The combined sample for the I/B/E/S Compustat merger covers the period 2002–2022 for 12,424 firm-years spanning 2,472 US firms. To construct our sample, we follow convention and discard utilities and financials, as well as any firm-years that have negative or non-existing values for revenue, employment, and/or the capital stock. We focus on revenue forecasts, which comprise the lion-share of all forecasts provided by managers, and analyze one-year ahead annual forecasts.⁷ We only use “centered forecasts”: that is, either point estimates or forecasts that are stated as a range. In the latter case, we use the mid-point of the range as the point estimate. We remove observations that are related to the top and bottom one percent of the forecast error distribution.

⁷For an individual firm, we study the first forecast made in the year (Jan–April) that pertains to the firm’s end-of-year financial results. Firms mainly report previous year’s financial results in Q1 of the following year.

Variable Definitions: We use the following variables from Compustat Fundamentals Annual: revenue (code: `sale`), profits (code: `ib`), capital (code: `ppent`), investment (code: `capx`), assets (code: `at`), employment (code: `emp`), and industry classification (code: `naics` and `sic`). We measure a firm’s debt as the total net value of liabilities (code: `d1lt+d1c-che`). Finally, Compustat only has limited data on firm wages. We use total labor and related expenses as our measure of the overall wage bill (code: `x1r`). We link the Compustat data with the I/B/E/S database using the CRSP ID that is available for both. The annual forecasts employed from the I/B/E/S managerial guidance database (code: `val1` and `val2`) are those that pertain to “centered forecasts” (code: `fdesc=1,2`) in millions or billions of USD. We mainly study forecasts of future revenue (code: `measure=SAL`), although we also consider forecasts of future profits (code: `measure=NET`) and earnings per share (code: `measure=EPS`). We define a firm’s forecast error of a variable as the difference between the realized value of said variable from Compustat and the one-year-ahead forecast of the variable from I/B/E/S.

Descriptive Statistics: We report descriptive statistics for our data set in Table A.15.

Table A.15: Descriptive Statistics: I/B/E/S-Compustat

Variable Name	Obs.	Mean	Std.	Median
Revenue	12,423	3,843	11,721	771.98
Profits	12,423	291.83	1,422	30.00
Capital	12,332	1,068	4,747	121.73
Investment	12,416	188.10	873	27.23
Wages	807	1,895	4,046	358.80
Debt	12,378	1,277	9,016	43.03
Assets	12,423	5,841	22,558	995.51
Employment	12,341	13.17	32.93	2.93
Revenue/capital	12,328	13.33	32.29	7.01
Forecast	12,328	14.81	51.83	7.04
Forecast Log	12,328	1.95	1.13	1.95
Forecast Error	12,084	-0.13	1.91	0.01
Forecast Error Log	12,080	-0.01	0.13	0.00

Notes: The table reports descriptive statistics for the sample of 2,472 firms from 2002-2022 in the combined Compustat-I/B/E/S database. The units of the first seven rows are USD millions. The employment row is in ’000-employees. The first eight rows capture, respectively, firm revenue, GAAP net-profits, book value of the capital stock, total value of capital expenditures, end-of-period total liabilities and assets, overall expenditures to labor and related expenses, and the total number of employees. The next three rows measure revenue scaled by a firm’s tangible capital and the (log) of the year-ahead forecast. The final two rows are for the year-ahead forecast error defined as realized future (log)-revenue minus (log of) the forecast. In the final two rows, observations have been removed that are in top and bottom one percent of the error distribution.

A.3 Data Construction: Duke CFO Survey

The CFO Survey is a quarterly survey of U.S. financial professionals designed to elicit the financial outlook for their firms, the challenges they face, and their expectations for the U.S. economy. We exploit a combination of the CFO Survey and Compustat Fundamentals Annual. The combined sample merger covers the period 2002-2011, the period for which the merger with Compustat is possible, for 638 firm-years spanning 325 US firms. In the CFO Survey sample, we remove forecasts that are not one-year-ahead forecasts of the fiscal year, as well as any firm-years that have non-existing profit forecasts. We focus on profits forecasts as these comprise the lion-share of all forecasts provided. The treatment of the Compustat data is described above. Firm size is measured by firm employment: 1 (fewer than 100 employees); 2 (100-999 employees); 3 (1,000-4,999 employees); and 4 (>4,999 employees).

B Proofs and derivations

Proof of Proposition 1: The proposition has two parts.

Part (i): The optimality conditions for labor demand and supply are, respectively:

$$\alpha \cdot \frac{\theta - 1}{\theta} \cdot \frac{\mathbb{E}[p_i \cdot y_i | \mu_i, s_i, \tau_i]}{l_i} = w \quad \text{and} \quad w = v'(L).$$

Thus,

$$l_i = \alpha \cdot \frac{\theta - 1}{\theta} \cdot \frac{\mathbb{E}[p_i \cdot y_i | \mu_i, s_i, \tau_i]}{v'(L)} = \alpha \cdot \frac{\theta - 1}{\theta} \cdot \frac{\mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i\right] \cdot l_i^{\alpha \cdot \frac{\theta-1}{\theta}} \cdot C^{\frac{1}{\theta}}}{v'(L)}.$$

We conclude that

$$l_i = \left(\mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i\right] \cdot \alpha \cdot \frac{\theta}{\theta - 1} \cdot C^{\frac{1}{\theta}} \cdot v'(L)^{-1} \right)^{\frac{1}{1 - \alpha \cdot \frac{\theta-1}{\theta}}} = \left(\mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i\right] \cdot \Omega \right)^{\frac{1}{1 - \alpha \cdot \frac{\theta-1}{\theta}}}. \quad (\text{A1})$$

Part (ii): Ex-ante expected profits for firm i with mean-productivity μ_i equals:

$$\mathbb{E}[\hat{\pi}_i | \mu_i, \tau_i] = \mathbb{E}[p_i \cdot y_i | \mu_i, \tau_i] - w \cdot \mathbb{E}[l_i | \mu_i, \tau_i] = \left(\frac{1 - \alpha \cdot \frac{\theta-1}{\theta}}{\alpha \cdot \frac{\theta-1}{\theta}} \right) \cdot w \cdot \mathbb{E}[l_i | \mu_i, \tau_i].$$

Thus, firm i with mean-productivity μ_i chooses to acquire or not depending on whether:

$$\left(\frac{1 - \alpha \cdot \frac{\theta-1}{\theta}}{\alpha \cdot \frac{\theta-1}{\theta}} \right) \cdot (\mathbb{E}[l_i | \mu_i, \tau_i = \bar{\tau}] - \mathbb{E}[l_i | \mu_i, \tau_i = \underline{\tau}]) \geq \chi.$$

Finally, it follows from (14) that

$$\begin{aligned}
\mathbb{E}[l_i | \mu_i, \tau_i] &= \mathbb{E} \left[\mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} | \mu_i, \tau_i \right] \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \\
&= \mathbb{E} \left[\left(\exp^{\frac{\theta-1}{\theta} \cdot \left(\frac{\tau_i}{\tau_i + \tau_a} \cdot s_i + \frac{\tau_a}{\tau_i + \tau_a} \cdot \mu_i \right) + \frac{1}{2} \cdot \left(\frac{\theta-1}{\theta} \right)^2 \cdot \frac{1}{\tau_i + \tau_a}} \right)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} | \mu_i, \tau_i \right] \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \\
&= \exp^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}} \cdot \frac{\theta-1}{\theta} \cdot \mu_i} \cdot \exp^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}} \cdot \frac{1}{2} \cdot \left(\frac{\theta-1}{\theta} \right)^2 \cdot \frac{\tau_a + \frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}} \cdot \tau_i}{\tau_i + \tau_a} \cdot \frac{1}{\tau_a}} \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}}
\end{aligned}$$

We conclude that

$$\mathbb{E}[l_i | \mu_i, \tau_i] = \left(\exp^{\frac{\theta-1}{\theta} \cdot \mu_i} \cdot g(\tau_i) \cdot \Omega \right)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}}.$$

□

Proof of Corollary ??: The (log-)total revenue productivity of firm i equals

$$\text{tfpr}_i = \log p_i y_i - \log l_i = \log A_i^{\frac{\theta-1}{\theta}} + \log C^{\frac{1}{\theta}} + \left(\alpha \frac{\theta-1}{\theta} - 1 \right) \log l_i, \quad (\text{A2})$$

while its forecast error is

$$\text{error}_i = \log(p_i \cdot y_i) - \log(\mathbb{E}[p_i \cdot y_i | \mu_i, s_i, \tau_i]) = \log A_i^{\frac{\theta-1}{\theta}} - \log \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]. \quad (\text{A3})$$

The result then follows from taking the conditional variance of (A2) and (A3)

□

Proof of Corollary ??: Let μ^* denote the threshold for which firm i is indifferent between acquiring information or not. It follows from the above that

$$\begin{aligned}
\left(\frac{1 - \alpha \cdot \frac{\theta-1}{\theta}}{\alpha \cdot \frac{\theta-1}{\theta}} \right) \cdot (\mathbb{E}[l_i | \mu_i = \mu^*, \tau_i = \bar{\tau}] - \mathbb{E}[l_i | \mu_i = \mu^*, \tau_i = \underline{\tau}]) &= \chi \\
\left(\exp^{\frac{\theta-1}{\theta} \cdot \mu^*} \cdot \Omega \right)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot \left[g(\bar{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} - g(\underline{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right] &= \left(\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1 - \alpha \cdot \frac{\theta-1}{\theta}} \right) \cdot \chi.
\end{aligned}$$

Thus,

$$\mu^*(\Omega) = \left(\frac{\theta}{\theta-1} - \alpha \right) \cdot \log \frac{\left(\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1 - \alpha \cdot \frac{\theta-1}{\theta}} \right) \cdot \chi}{\left[g(\bar{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} - g(\underline{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right] \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}}}.$$

□

Proof of Proposition ??: The equilibrium of the model is characterized by the equilibrium in the labor market and the goods market. The *labor market clearing condition* is:

$$\begin{aligned} L &= N \cdot (1 - G(\mu^*)) \cdot \chi + \int_0^N l_i di \\ &= N \cdot \Phi\left(\frac{\mu - \mu^*}{\tau_\mu^{-1/2}}\right) \cdot \chi + \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot \int_0^N \mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | s_i, \tau\right]^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} di. \end{aligned}$$

Recall that $\gamma = \frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}$. Splitting the firms into those that acquire information ($\mu < \mu^*$) and those that do not ($\mu > \mu^*$), a few simple but tedious derivations show that⁸:

$$\int_0^N \mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | s_i, \tau\right]^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} = N \cdot \left[\Upsilon(\bar{\tau}) + [\Upsilon(\underline{\tau}) - \Upsilon(\bar{\tau})] \cdot \Phi\left(\frac{(\mu^* - \mu) - \gamma \cdot \frac{\theta-1}{\theta} \cdot \frac{1}{\tau_\mu}}{\tau_\mu^{-1/2}}\right) \right],$$

where

$$\Upsilon(\tau_i) \equiv \exp^{\gamma \cdot \frac{\theta-1}{\theta} \cdot \mu + \frac{1}{2} \cdot \gamma \cdot \left(\frac{\theta-1}{\theta}\right)^2 \cdot \left[\frac{1}{\tau_a + \tau_i} + \frac{\gamma}{\tau_\mu} + \gamma \frac{\tau_i}{\tau_a + \tau_i} \cdot \frac{1}{\tau_a}\right]} = \Lambda \cdot g(\tau_i)^\gamma,$$

and $\Lambda \equiv \exp^{\gamma \cdot \frac{\theta-1}{\theta} \cdot \mu + \frac{1}{2} \cdot \gamma \cdot \left(\frac{\theta-1}{\theta}\right)^2 \cdot \frac{1}{\tau_\mu}}$. Thus, using that $z^* = (\mu^* - \mu) \cdot \tau_\mu^{1/2}$, we find that

$$L = N \cdot [\Phi(-z^*) \cdot \chi + \Gamma(z^*) \cdot \Omega^\gamma], \quad (\text{A4})$$

where

$$\begin{aligned} \Gamma(z^*) &\equiv \frac{1}{N} \cdot \int_0^N \mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | s_i, \tau\right]^\gamma di \\ &= \Lambda \cdot \left[g(\bar{\tau})^\gamma + [g(\underline{\tau})^\gamma - g(\bar{\tau})^\gamma] \cdot \Phi\left(z^* - \gamma \cdot \frac{\theta-1}{\theta} \cdot \tau_\mu^{-\frac{1}{2}}\right) \right]. \end{aligned} \quad (\text{A5})$$

The *goods market clearing condition* is:

$$\begin{aligned} C &= \left(\int_0^N A_i^{\frac{\theta-1}{\theta}} \cdot l_i^{\alpha \cdot \frac{\theta-1}{\theta}} \cdot di \right)^{\frac{\theta}{\theta-1}} = \left(\int_0^N A_i^{\frac{\theta-1}{\theta}} \cdot \left(\mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i\right] \cdot \Omega \right)^{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot di \right)^{\frac{\theta}{\theta-1}} \\ &= \left(\int_0^N A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E}\left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i\right]^{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot di \right)^{\frac{\theta}{\theta-1}} \cdot \Omega^{\frac{\alpha}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \end{aligned}$$

⁸The derivations require computing the conditional mean of a log-normal. Suppose $X \sim \mathcal{LN}(\mu, \sigma^2)$. Then, $\mathbb{E}[X | X < k] = \exp^{\mu + \frac{\sigma^2}{2}} \cdot \frac{\Phi\left[\frac{\log k - \mu - \sigma^2}{\sigma}\right]}{\Phi\left[\frac{\log k - \mu}{\sigma}\right]}$ and $\mathbb{E}[X | X > k] = \exp^{\mu + \frac{\sigma^2}{2}} \cdot \frac{\Phi\left[\frac{-\log k - \mu - \sigma^2}{\sigma}\right]}{1 - \Phi\left[\frac{\log k - \mu}{\sigma}\right]}$.

Now, notice that

$$\begin{aligned}
\int_0^N A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot di &= N \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right] \\
&= N \cdot \mathbb{E}_{\mu_i, s_i, \tau_i} \left\{ \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \mid \mu_i, s_i, \tau_i \right] \right\} \\
&= N \cdot \mathbb{E}_{\mu_i, s_i, \tau_i} \left[\mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\gamma \right] = \int_0^N \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, s_i, \tau_i \right]^\gamma di,
\end{aligned}$$

where the first and last equality follow from the *Law of Large Numbers*, and $\mathbb{E}_x[\cdot]$ denotes the expectation taken with respect to the random variable x . We conclude that:

$$C = [N \cdot \Gamma(z^*)]^{\frac{\theta}{\theta-1}} \cdot \Omega^{\frac{\alpha}{1-\alpha \cdot \frac{\theta-1}{\theta}}}. \quad (\text{A6})$$

Re-arranging equations (A4) and (A6) show that

$$\mathcal{L} \equiv L - N \cdot \Phi(-z^*) \cdot \chi = N \cdot \Gamma(z^*) \cdot \Omega^\gamma$$

and that

$$C = \mathcal{A}(z^*) \cdot \mathcal{L}(z^*)^\alpha, \quad \mathcal{A}(z^*) = [N \cdot \Gamma(z^*)]^{\frac{\theta}{\theta-1}-\alpha}.$$

Substituting out $\Gamma(z^*)$ then completes the proof. $\square$

Proof of Corollary 2: The equilibrium cut-off z^* , associated with μ^* , is given by the intersection of two schedules: (i) *the information choice schedule*

$$\mu + \tau_\mu^{-1/2} \cdot z^* = \left(\frac{\theta}{\theta-1} - \alpha \right) \cdot \log \frac{\frac{\alpha \cdot \frac{\theta-1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}} \cdot \chi}{\left[g(\bar{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} - g(\underline{\tau})^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right] \cdot \Omega(z^*)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}}} \quad (\text{A7})$$

and (ii) *the general equilibrium schedule*, which combines the definition of Ω in Definition 2 with (??):

$$\Omega(z^*) = \frac{\alpha \cdot \frac{\theta-1}{\theta} \cdot \left(N^{\frac{1}{\theta-1}} \cdot \Gamma(z^*)^{\frac{1}{\theta-1}} \cdot \Omega(z^*)^{\frac{\alpha \cdot \frac{1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right)}{v' \left(N \cdot \left[\Phi(-z^*) \cdot \chi + \Gamma(z^*) \cdot \Omega(z^*)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right] \right)}. \quad (\text{A8})$$

We here show that an intersection between the two schedules always exists.

Schedule (i): the information choice schedule. Notice that the schedule in (A7) is continuous

in (z^*, Ω) and has the characteristic that $\partial\Omega/\partial z^* < 0$, where $\Omega > 0$. We also have that $\lim_{z \rightarrow -\infty} \Omega_{z \rightarrow -\infty} = +\infty$ and $\lim_{z \rightarrow +\infty} \Omega_{z \rightarrow +\infty} = 0$.

Schedule (ii): the general equilibrium schedule. Notice that the schedule in (A7) is continuous

in (z^*, Ω) and has the characteristics that $\lim_{z \rightarrow +\infty} \Omega_{z \rightarrow +\infty} = \frac{\alpha \cdot \frac{\theta-1}{\theta} \cdot \left(N^{\frac{1}{\theta-1}} \cdot \underline{\Gamma}^{\frac{1}{\theta-1}} \cdot \Omega^{\frac{\alpha \cdot \frac{1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right)}{v' \left(N \cdot \underline{\Gamma} \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right)} \in (0, \infty)$,

where $\underline{\Gamma} \equiv \mathcal{B} \cdot g(\underline{\tau})^\gamma$, and $\lim_{z \rightarrow -\infty} \Omega_{z \rightarrow -\infty} = \frac{\alpha \cdot \frac{\theta-1}{\theta} \cdot \left(N^{\frac{1}{\theta-1}} \cdot \bar{\Gamma}^{\frac{1}{\theta-1}} \cdot \Omega^{\frac{\alpha \cdot \frac{1}{\theta}}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right)}{v' \left(N \cdot \bar{\Gamma} \cdot \Omega^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \right)} \in (0, \infty)$ with $\bar{\Gamma} \equiv \mathcal{B} \cdot g(\bar{\tau})^\gamma$.

The continuity of both schedules, and the fact the general equilibrium schedule admits finite values for all z , ensures that at least one intersection exists between the two schedules. $\square$

Proof of Corollary ??: The PE-schedule for the marginal firm acquiring information in (20) can be re-written as:

$$\Omega_{PE}^\gamma(z) = \exp^{\kappa(\chi) - \gamma \cdot \frac{\theta-1}{\theta} \cdot (\mu + \tau_\mu^{-0.5} \cdot z)}, \quad \kappa(\chi) \equiv \log \frac{\alpha \cdot \gamma \cdot \frac{\theta-1}{\theta} \cdot \chi}{[g(\bar{\tau})^\gamma - g(\underline{\tau})^\gamma]}. \quad (\text{A9})$$

Notice that $\partial\Omega_{PE}^\gamma/\partial z < 0$, $\partial^2\Omega_{PE}^\gamma/\partial z^2 > 0$, and that $\lim_{z \rightarrow +\infty} \Omega_{PE}^\gamma = 0$ and $\lim_{z \rightarrow -\infty} \Omega_{PE}^\gamma = \infty$. The GE-schedule in (??), in contrast, equals:

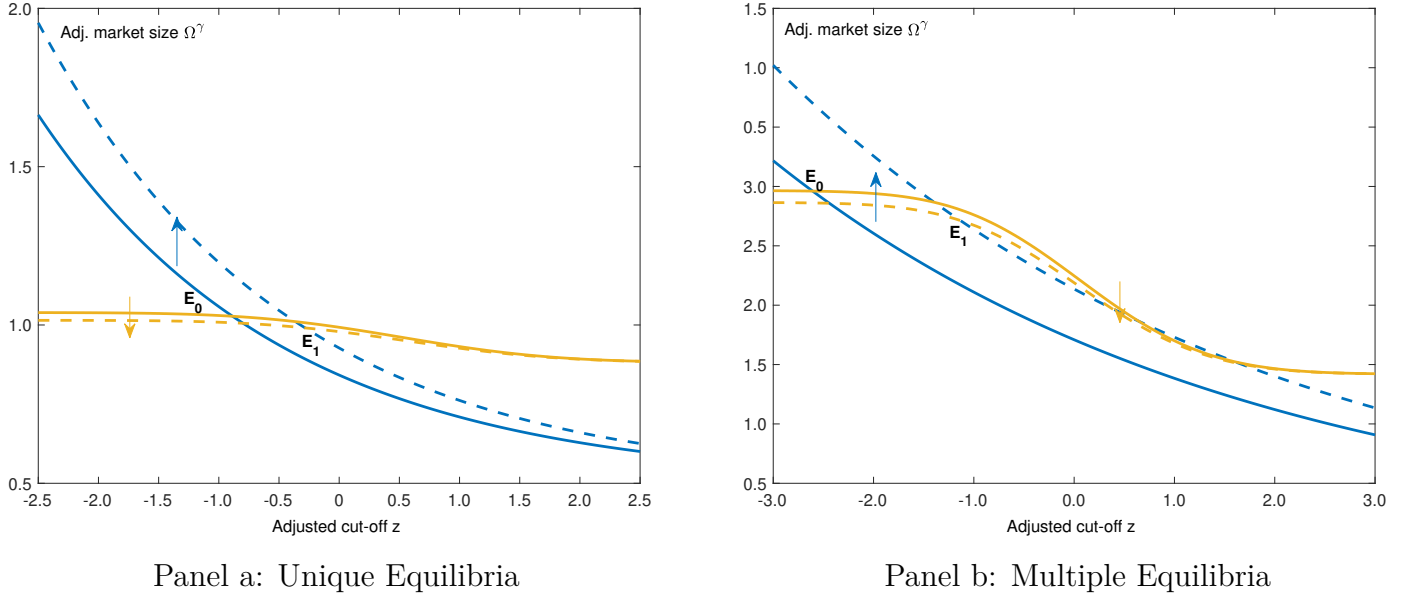
$$\Omega_{GE}^\gamma(z) = \left[\frac{v' \left(N \cdot \Phi(-z) \cdot \chi + N \cdot \Gamma(z) \cdot \Omega^\gamma \right)}{\alpha \cdot \frac{\theta-1}{\theta} \cdot N^{\frac{1}{\theta-1}} \cdot \Gamma(z)^{\frac{1}{\theta-1}}} \right]^{\frac{1}{\alpha-1}}, \quad (\text{A10})$$

which can be either increasing and decreasing in z^* . Notice that both $\lim_{z \rightarrow -\infty} \Omega_{GE}^\gamma$ and $\lim_{z \rightarrow +\infty} \Omega_{GE}^\gamma$ admit (finite) values in $(0, \infty)$.

An equilibrium $(z, \Omega^\gamma)^*$ corresponds to a crossing between the two schedules.

Because $\lim_{z \rightarrow -\infty} \Omega_{GE}^\gamma \in (0, \infty)$, while $\lim_{z \rightarrow -\infty} \Omega_{PE}^\gamma = \infty$, the minimum-value for z^* has to occur at a point in which the GE-schedule crosses the PE-schedule from below in (z, Ω^γ) -space. Figure xx provides several examples. At this equilibrium point, a marginal *increase* in χ moves up the PE-schedule, but down the GE-schedule. The latter follows from $v'(\cdot) > 0$ and $\alpha \in (0, 1)$. As a result, the minimum-value of z^* increases, irrespective of whether the GE-schedule is upward- or downward-sloping, or how many equilibria there are. $\square$

Figure B.1: Comparative Statics of an Increase in χ



Note: Dashed lines correspond to after the increase in χ ; solid lines are before the increase.

Proof of Proposition 4: The *Law of Large Numbers* implies that:

$$C = \left(\int_0^N A_i^{\frac{\theta-1}{\theta}} \cdot l_i(\mu_i, \tau_i, s_i)^{\alpha \cdot \frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}} = \left(\int_0^N \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot l_i(\mu_i, \tau_i, s_i)^{\alpha \cdot \frac{\theta-1}{\theta}} | \mu_i, \tau_i, s_i \right] \cdot di \right)^{\frac{\theta}{\theta-1}}.$$

Thus, letting $\lambda \in \mathbb{R}$ denote the multiplier on the feasibility constraint (i.e., the shadow wage), the social planner's first order conditions w.r.t., L and $\{l_i\}$ yield, respectively:

$$v'(L) = \lambda, \quad \text{and} \quad \alpha \cdot C^{\frac{1}{\theta}} \cdot \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot l_i^{\alpha \cdot \frac{\theta-1}{\theta} - 1} | \mu_i, \tau_i, s_i \right] = \lambda. \quad (\text{A11})$$

The planner, furthermore, makes firm i produce information, i.e., $l_i^* = 1$, if and only if:

$$\frac{\theta}{\theta-1} \cdot C^{\frac{1}{\theta}} \cdot \left(\mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot l_i^{\alpha \cdot \frac{\theta-1}{\theta}} | \mu_i, \tau_i = \bar{\tau} \right] - \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} \cdot l_i^{\alpha \cdot \frac{\theta-1}{\theta}} | \mu_i, \tau_i = \underline{\tau} \right] \right) \geq \lambda \cdot (\chi + \mathbb{E} [l_i | \mu_i, \tau_i = \bar{\tau}] - \mathbb{E} [l_i | \mu_i, \tau_i = \underline{\tau}]) \quad (\text{A12})$$

Combined, equations (A11) and (A12) pin down efficient employment l_i^* and information production l_i^* as a function of the efficient market size $\hat{\Omega} \equiv \alpha \cdot \left(\frac{C^{\frac{1}{\theta}}}{v'(L)} \right)^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}}$ of the economy. Specifically, we find that:

$$l_i^* = \mathbb{E} \left[A_i^{\frac{\theta-1}{\theta}} | \mu_i, \tau_i, s_i \right]^{\frac{1}{1-\alpha \cdot \frac{\theta-1}{\theta}}} \cdot \hat{\Omega},$$

and

$$\iota_i^* = \begin{cases} 1 & \text{if } \frac{1-\frac{\theta-1}{\theta} \cdot \alpha}{\frac{\theta-1}{\theta} \cdot \alpha} \cdot (\mathbb{E}[\iota_i^* | \mu_i, \tau_i = \bar{\tau}] - \mathbb{E}[\iota_i^* | \mu_i, \tau_i = \underline{\tau}]) \geq \chi \\ 0 & \text{otherwise} \end{cases},$$

where the latter also pins down the multiplier $\mu^* \in \mathbb{R}$ on the information constraint. $\square$

Proof of Corollary 4: The ex-post profits of firm i with the labor subsidy $s \in (0, 1)$ equal:

$$\pi_i = p_i \cdot y_i - (1 - s) \cdot w \cdot (l_i + \chi \cdot \iota_i).$$

The first order condition for the choice of l_i is thus given by:

$$\alpha \cdot \frac{\theta - 1}{\theta} \cdot \frac{\mathbb{E}[p_i \cdot y_i | \mu_i, s_i, \tau_i]}{l_i} = (1 - s) \cdot w,$$

which immediately arrives us at our desired result. $\square$

References

- ANGELETOS, G.-M., HUO, Z. and SASTRY, K. A. (2021). Imperfect macroeconomic expectations: Evidence and theory. *NBER Macroeconomics Annual*, **35** (1), 1–86.
- and LIAN, C. (2016). Incomplete information in macroeconomics: Accommodating frictions in coordination. *Handbook of Macroeconomics*, **2**, 1065–1240.
- and PAVAN, A. (2007). Efficient Use of Information and Social Value of Information. *Econometrica*, **75** (4), 1103–1142.
- AUTOR, D., DORN, D., KATZ, L. F., PATTERSON, C. and VAN REENEN, J. (2020). The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics*, **135** (2), 645–709.
- BALEY, I. and VELDKAMP, L. (2025). *The Data Economy: Tools and Applications*. Princeton University Press.
- BLANCHARD, O. J. and KİYOTAKI, N. (1987). Monopolistic competition and the effects of aggregate demand. *The American Economic Review*, pp. 647–666.
- , L’HUILIER, J.-P. and LORENZONI, G. (2013). News, noise, and fluctuations: An empirical exploration. *American Economic Review*, **103** (7), 3045–70.
- BLOOM, N., BRYNJOLFSSON, E., FOSTER, L., JARMIN, R., PATNAIK, M., SAPORTA-EKSTEN, I. and VAN REENEN, J. (2019). What drives differences in management practices? *American Economic Review*, **109** (5).
- BRYNJOLFSSON, E., MCAFEE, A., SORELL, M. and ZHU, F. (2008). Scale without mass:

- business process replication and industry dynamics. *Harvard Business School Technology & Operations Mgt. Unit Research Paper*, (07-016).
- CHAHROUR, R. and JURADO, K. (2018). News or noise? the missing link. *American Economic Review*, **108** (7), 1702–36.
- CHIAVARI, A. and GORAYA, S. (2023). The rise of intangible capital and the macroeconomic implicationsn. *mimeo*.
- DAVID, J. M., HOPENHAYN, H. A. and VENKATESWARAN, V. (2016). Information, misallocation, and aggregate productivity. *The Quarterly Journal of Economics*, **131** (2), 943–1005.
- and VENKATESWARAN, V. (2019). The sources of capital misallocation. *American Economic Review*, **109** (7), 2531–2567.
- EDMOND, C., MIDRIGAN, V. and XU, D. Y. (2023). How costly are markups? *Journal of Political Economy*, **131** (7), 000–000.
- FOSTER, L., HALTIWANGER, J. and SYVERSON, C. (2008). Reallocation, firm turnover, and efficiency: Selection on productivity or profitability? *American Economic Review*, **98** (1), 394–425.
- GOPINATH, G., KALEMLI-ÖZCAN, Ş., KARABARBOUNIS, L. and VILLEGAS-SANCHEZ, C. (2017). Capital allocation and productivity in south europe. *The Quarterly Journal of Economics*, **132** (4), 1915–1967.
- GRAHAM, J., MEYER, B., PARKER, N. and WADDEL, S. (2023). The cfo survey. *Duke University mimeo*.
- HSIEH, C.-T. and KLENOW, P. J. (2009). Misallocation and manufacturing tfp in china and india. *The Quarterly journal of economics*, **124** (4), 1403–1448.
- LORENZONI, G. (2009). A theory of demand shocks. *American Economic Review*, **99** (5), 2050–84.
- LUCAS, R. E. J. (1972). Expectations and the neutrality of money. *Journal of Economic Theory*, **4** (2), 103–124.
- MCKINSEY and CO. (2023). Frontiers of the data economy?
- MYATT, D. P. and WALLACE, C. (2011). Endogenous information acquisition in coordination games. *The Review of Economic Studies*, **79** (1), 340–374.
- OTTONELLO, P. and WINBERRY, T. (2020). Financial heterogeneity and the investment channel of monetary policy. *Econometrica*, **88** (6), 2473–2502.